BUSINESS REPORT

Health Care Capstone Project

By

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1. Business Problem

We all know that Health care is very important domain in the market. It is directly linked with the life of the individual; hence we must be always be proactive in this domain. Money plays a major role in this domain, because sometime treatment becomes super costly and if any individual is not covered under the insurance, then it will become a tough financial situation for that individual. The companies in the medical insurance also want to reduce their risk by optimizing the insurance cost, because we all know a healthy body is in the hand of the individual only. If individual eat healthy and do proper exercise the chance of getting ill is drastically reduced.

2. Goal & Objective

The objective of this exercise is to build a model, using data that provide the optimum insurance cost for an individual. You must use the health and habit related parameters for the estimated cost of insurance

3. Data Dictionary

- **applicant_id:** Applicant unique ID
- **years_of_insurance_with_us:** Since how many years customer is taking policy from the same company only
- **regular_checkup_lasy_year:** Number of times customers has done the regular health check-up in last one year
- **adventure_sports:** Customer is involved with adventure sports like climbing, diving etc.
- **Occupation:** Occupation of the customer
- **visited_doctor_last_1_year:** Number of times customer has visited doctor in last one year
- **Cholesterol_level:** Cholesterol level of the customers while applying for insurance
- **daily_avg_steps:** Average daily steps walked by customers
- **age:** Age of the customer
- **heart_decs_history:** Any past heart diseases
- **other_major_decs_history:** Any past major diseases apart from heart like any operation
- **Gender:** Gender of the customer
- **avg_glucose_level:** Average glucose level of the customer while applying the insurance
- **bmi:** BMI of the customer while applying the insurance
- **smoking_status:** Smoking status of the customer
- **Year_last_admitted:** When customer have been admitted in the hospital last time
- **Location:** Location of the hospital
- **weight:** Weight of the customer

- **covered_by_any_other_company:** Customer is covered from any other insurance company
- **Alcohol:** Alcohol consumption status of the customer
- **exercise:** Regular exercise status of the customer
- **weight_change_in_last_one_year:** How much variation has been seen in the weight of the customer in last year
- **fat_percentage:** Fat percentage of the customer while applying the insurance
- **insurance_cost:** Total Insurance cost

4. Data Overview through Python

Data which is stored in csv file is imported to Jupyter Notebook with the help of pandas, NumPy libraries and used matplotlib and seaborn libraries to help with data visualization. Dataset is stored in python variable named data.

	applicant_id	years_of_insurance_with_us	regular_checkup_lasy_year	adventure_sports	Occupation	visited_doctor_last_1_year	cholesterol_level	daily_avg_steps	age
0	5000	3	1	1	Salried	2	125 to 150	4866	28
1	5001	0	0	0	Student	4	150 to 175	6411	50
2	5002	1	0	0	Business	4	200 to 225	4509	68
3	5003	7	4	0	Business	2	175 to 200	6214	51
4	5004	3	1	0	Student	2	150 to 175	4938	44

Figure 1: Data Head

	applicant_id	years_of_insurance_with_us	regular_checkup_lasy_year	adventure_sports	Occupation	visited_doctor_last_1_year	cholesterol_level	daily_avg_steps
24995	29995	3	0	0	Salried	4	225 to 250	5614
24996	29996	6	0	0	Business	4	200 to 225	4719
24997	29997	7	0	1	Student	2	150 to 175	5624
24998	29998	1	0	0	Salried	2	225 to 250	10777
24999	29999	8	2	0	Business	4	150 to 175	5882

Figure 2: Data Tail

The dataset has **25000** rows and **24** columns.

Checking the data types of data.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25000 entries, 0 to 24999
Data columns (total 24 columns):
 #   Column                          Non-Null Count  Dtype
---  ------                          --------------  -----
 0   applicant_id                    25000 non-null  int64
 1   years_of_insurance_with_us      25000 non-null  int64
 2   regular_checkup_lasy_year       25000 non-null  int64
 3   adventure_sports                25000 non-null  int64
 4   Occupation                      25000 non-null  object
 5   visited_doctor_last_1_year      25000 non-null  int64
 6   cholesterol_level               25000 non-null  object
 7   daily_avg_steps                 25000 non-null  int64
 8   age                             25000 non-null  int64
 9   heart_decs_history              25000 non-null  int64
 10  other_major_decs_history        25000 non-null  int64
 11  Gender                          25000 non-null  object
 12  avg_glucose_level               25000 non-null  int64
 13  bmi                             24010 non-null  float64
 14  smoking_status                  25000 non-null  object
 15  Year_last_admitted              13119 non-null  float64
 16  Location                        25000 non-null  object
 17  weight                          25000 non-null  int64
 18  covered_by_any_other_company    25000 non-null  object
 19  Alcohol                         25000 non-null  object
 20  exercise                        25000 non-null  object
 21  weight_change_in_last_one_year  25000 non-null  int64
 22  fat_percentage                  25000 non-null  int64
 23  insurance_cost                  25000 non-null  int64
dtypes: float64(2), int64(14), object(8)
memory usage: 4.6+ MB
```

Figure 3: Data Types

4.1 Checking and Treating Missing values in the data

```
applicant_id                        0
years_of_insurance_with_us          0
regular_checkup_lasy_year           0
adventure_sports                    0
Occupation                          0
visited_doctor_last_1_year          0
cholesterol_level                   0
daily_avg_steps                     0
age                                 0
heart_decs_history                  0
other_major_decs_history            0
Gender                              0
avg_glucose_level                   0
bmi                               990
smoking_status                      0
Year_last_admitted              11881
Location                            0
weight                              0
covered_by_any_other_company        0
Alcohol                             0
exercise                            0
weight_change_in_last_one_year      0
fat_percentage                      0
insurance_cost                      0
dtype: int64
```

- There are missing values in **BMI** and **Year_last_admitted** attributes

Figure 4: Missing data

4.2 Checking for duplicate values in the data.

```
0
```

- No duplicates in the dataset.

Dropping applicant_id attribute as it doesnot contribute any value to in Data understanding stage

4.3 Checking the statistical summary of the data.

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
years_of_insurance_with_us	25000.0	NaN	NaN	NaN	4.08904	2.606612	0.0	2.0	4.0	6.0	8.0
regular_checkup_lasy_year	25000.0	NaN	NaN	NaN	0.77368	1.199449	0.0	0.0	0.0	1.0	5.0
adventure_sports	25000.0	NaN	NaN	NaN	0.08172	0.273943	0.0	0.0	0.0	0.0	1.0
Occupation	25000	3	Student	10169	NaN	NaN	NaN	NaN	NaN	NaN	NaN
visited_doctor_last_1_year	25000.0	NaN	NaN	NaN	3.1042	1.141663	0.0	2.0	3.0	4.0	12.0
cholesterol_level	25000	5	150 to 175	8763	NaN	NaN	NaN	NaN	NaN	NaN	NaN
daily_avg_steps	25000.0	NaN	NaN	NaN	5215.88932	1053.179748	2034.0	4543.0	5089.0	5730.0	11255.0
age	25000.0	NaN	NaN	NaN	44.91832	16.107492	16.0	31.0	45.0	59.0	74.0
heart_decs_history	25000.0	NaN	NaN	NaN	0.05464	0.227281	0.0	0.0	0.0	0.0	1.0
other_major_decs_history	25000.0	NaN	NaN	NaN	0.09816	0.297537	0.0	0.0	0.0	0.0	1.0
Gender	25000	2	Male	16422	NaN	NaN	NaN	NaN	NaN	NaN	NaN
avg_glucose_level	25000.0	NaN	NaN	NaN	167.53	62.729712	57.0	113.0	168.0	222.0	277.0
bmi	24010.0	NaN	NaN	NaN	31.393328	7.876535	12.3	26.1	30.5	35.6	100.6
smoking_status	25000	4	never smoked	9249	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Year_last_admitted	13119.0	NaN	NaN	NaN	2003.892217	7.581521	1990.0	1997.0	2004.0	2010.0	2018.0
Location	25000	15	Bangalore	1742	NaN	NaN	NaN	NaN	NaN	NaN	NaN
weight	25000.0	NaN	NaN	NaN	71.61048	9.325183	52.0	64.0	72.0	78.0	96.0
covered_by_any_other_company	25000	2	N	17418	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Alcohol	25000	3	Rare	13752	NaN	NaN	NaN	NaN	NaN	NaN	NaN
exercise	25000	3	Moderate	14638	NaN	NaN	NaN	NaN	NaN	NaN	NaN
weight_change_in_last_one_year	25000.0	NaN	NaN	NaN	2.51796	1.690335	0.0	1.0	3.0	4.0	6.0
fat_percentage	25000.0	NaN	NaN	NaN	28.81228	8.632382	11.0	21.0	31.0	36.0	42.0
insurance_cost	25000.0	NaN	NaN	NaN	27147.40768	14323.691832	2468.0	16042.0	27148.0	37020.0	67870.0

Figure 5: Statistical Data summary

Observations:

years_of_insurance_with_us: Mean ~4 years. Indicates moderate loyalty. Longer duration means more loyalty

regular_checkup_lasy_year: Mean ~1.9 checkups. Reflects moderate proactive care; higher values may indicate stronger health engagement.

visited_doctor_last_1_year: Avg ~3.1 visits. Signals potential existing health concerns or better monitoring.

Cholesterol_level: Most frequent range: 150–175 mg/dL. Mid-range.

daily_avg_steps: Mean ~5,216. Moderately active.

age: Mean ~45 years.

Gender: Male majority.

avg_glucose_level: Mean ~168. High average—potential risk indicator for diabetes, affects premium calculation.

bmi: Mean ~31.4.

smoking_status: Dominated by “never smoked”.

Location: Could reflect regional cost variation.

weight: Mean ~84 kg.

covered_by_any_other_company: Mostly “No”

Alcohol: “Rare” is dominant. Low-risk group; could adjust pricing minimally based on category.

exercise: “Moderate” dominates. Positive lifestyle indicator—linked to reduced claims risk.

weight_change_in_last_one_year: Avg ~2.9 kg. Could indicate lifestyle change, recovery, or instability.

fat_percentage: Mean ~28.8%. Higher fat% aligns with elevated BMI—signals cardio-metabolic risk.

insurance_cost: Mean ~₹27,147.

5.0 Missing Value Imputation

- **BMI**

Imputing Mean value for the Group of Gender, Exercise, Age

- **Year_last_admitted**

The idea behind Year Last Admitted is to define what counts as a recent hospital admission.

Engineering this feature to **Recent Admission** would Flag admissions within certain Range like 2 or 3 years as clinically relevant.

But that again depends on assumption that missing values in this feature as never admitted which in turn makes this feature very synthetic.

Dropping this feature makes more sense.

- **applicant_id**

Dropping applicant_id attribute as it doesnot contribute any value to in Data understanding stage

5.1 Missing values after Imputation

```
years_of_insurance_with_us        0
regular_checkup_lasy_year         0
adventure_sports                  0
Occupation                        0
visited_doctor_last_1_year        0
cholesterol_level                 0
daily_avg_steps                   0
age                               0
heart_decs_history                0
other_major_decs_history          0
Gender                            0
avg_glucose_level                 0
bmi                               0
smoking_status                    0
Location                          0
weight                            0
covered_by_any_other_company      0
Alcohol                           0
exercise                          0
weight_change_in_last_one_year    0
fat_percentage                    0
insurance_cost                    0
dtype: int64
```

Figure 6: Missing data after imputation

6.0 Checking for Outliers

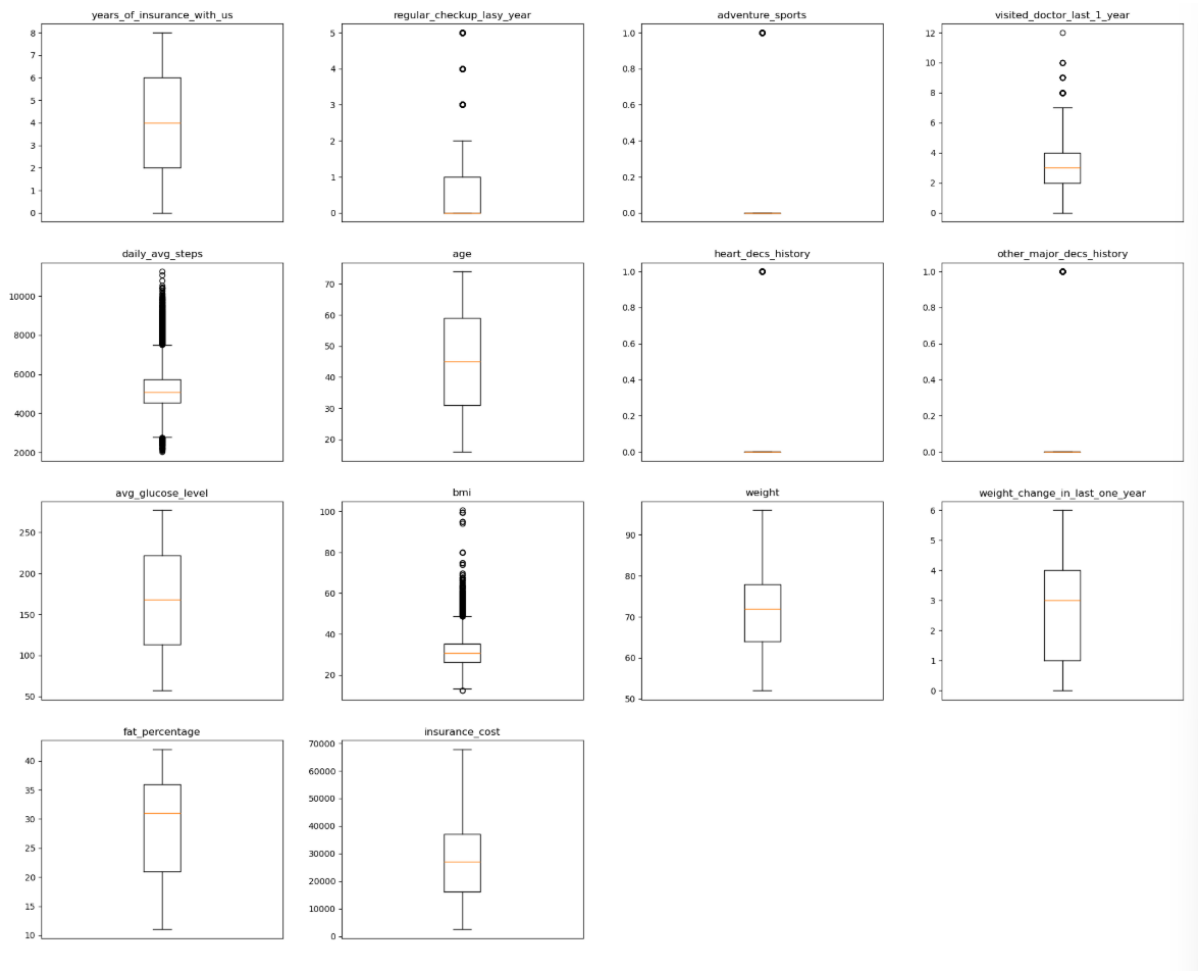


Figure 7: Outliers in data

All are real data not treating any outliers

6.1 Feature Engineering

To prepare categorical variables for modelling, As this data variables has some precedence the following mappings were applied:

- Cholesterol Level: Cholesterol ranges were converted to ordinal levels:
  1) **125–150 →** Level 0
  2) **150–175 →** Level 1
  3) **175–200 →** Level 2
  4) **200–225 →** Level 3
  5) **225–250 →** Level 4

- **Smoking Status:** Smoking categories were encoded as follows:
    1) **Never smoked → 0**
    2) **Formerly smoked → 1**
    3) **Smokes → 2**
    4) **Unknown → 3**
- **Exercise Intensity:** Exercise habits were encoded as:
    1) **Moderate → 0**
    2) **Extreme → 1**
    3) **No exercise → 2**
- **BMI**: for modelling and segmentation, the following banding strategy was applied:
    1) **BMI < 25** → Category **0** (typically considered *normal weight*)
    2) **25 ≤ BMI < 35** → Category **1** (typically *overweight to moderately obese*)
    3) **BMI ≥ 35** → Category **2** (typically *severely obese*)
- **fat percentage** values, the following banding logic was applied:
    1) **Fat % < 25** → Category **0** (*typically lean or healthy range*)
    2) **25 ≤ Fat % < 35** → Category **1** (*moderate fat range*)
    3) **Fat % ≥ 35** → Category **2** (*high fat range, potential health risk*)

7.0 Exploratory Data Analysis

7.1 Univariate Analysis

years_of_insurance_with_us

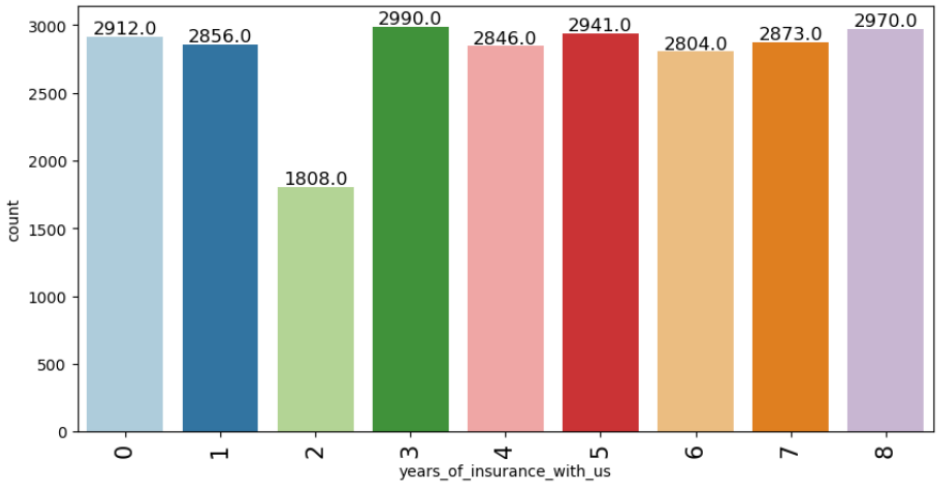


Figure 8: years_of_insurance_with_us

Observations:

- Peak at 3 Years: Highest concentration is at 3 years (2,990 customers)

regular_checkup_lasy_year

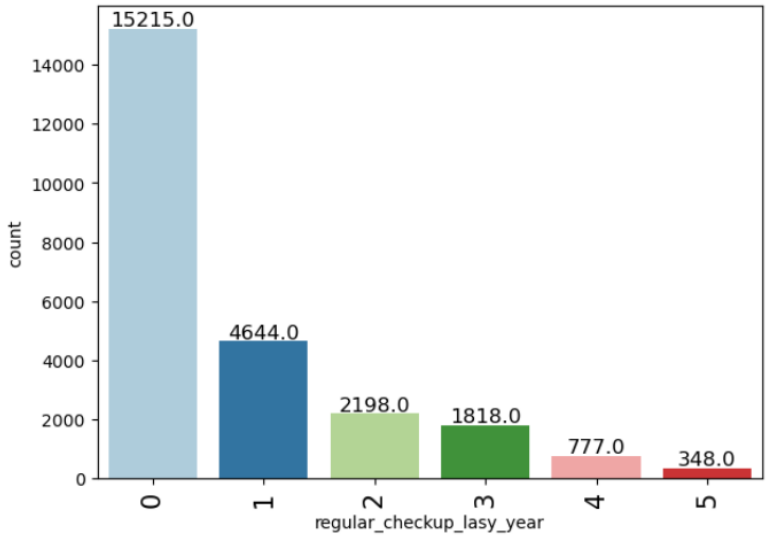


Figure 9: regular_checkup_lasy_year

Observations:

- Most customers do not engage in regular checkups.

adventure_sports

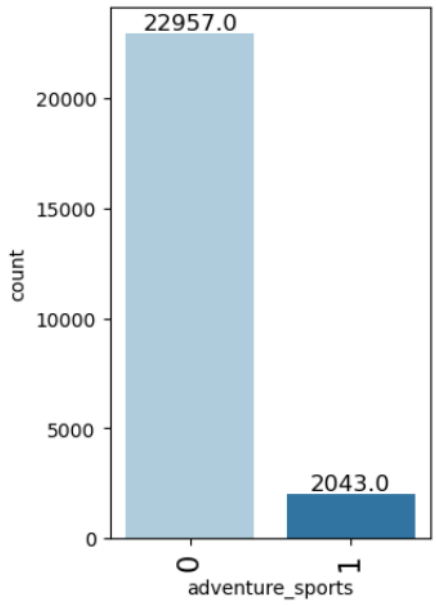


Figure 10: adventure_sports

Observations:

- Most customers do not engage in adventures sports.

Occupation

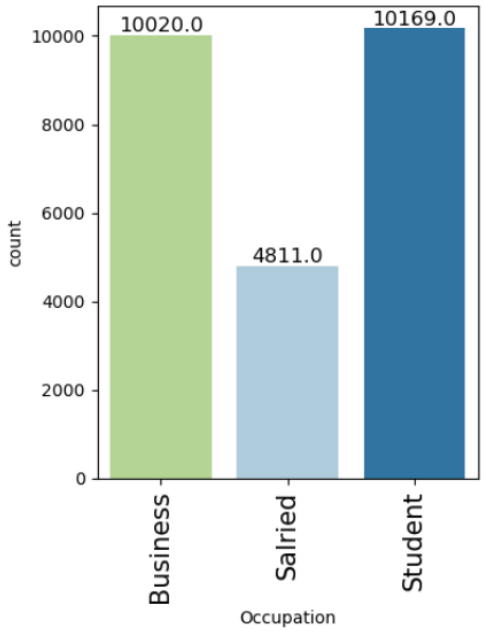


Figure 11: Occupation

Observations: Most of the data belongs to Business and Student occupation

visited_doctor_last_1_year

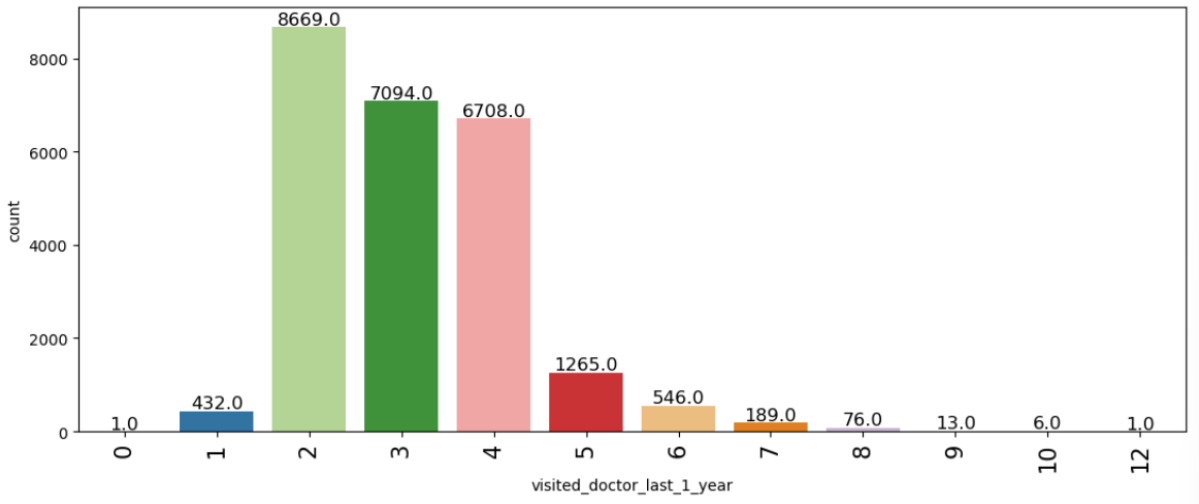


Figure 12: visited_doctor_last_1_year

Observations:

- Most customers visited 2,3.4 times last year

cholesterol_level

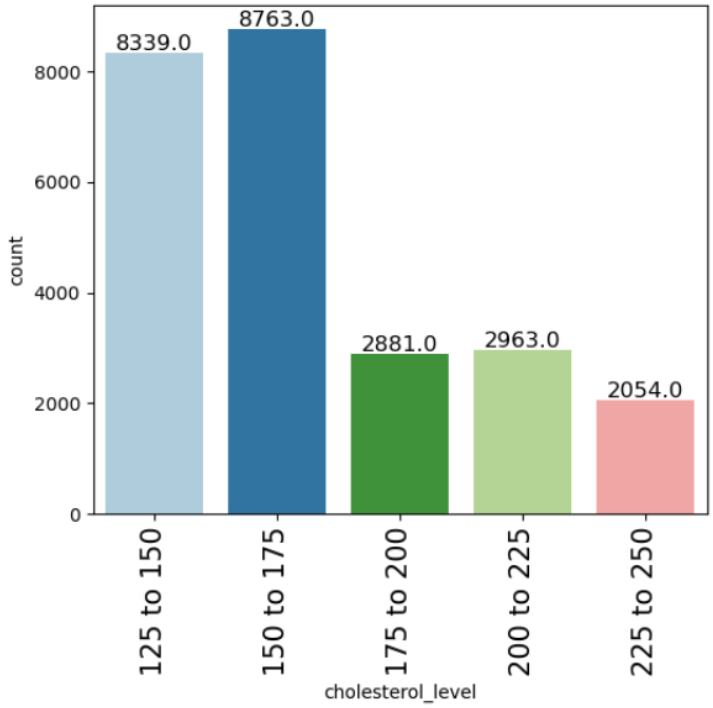


Figure 13: cholesterol_level

Observations:

- Majority of individuals fall within 125–175 range — relatively safe zone. Encouraging from a population health perspective.

daily_avg_steps

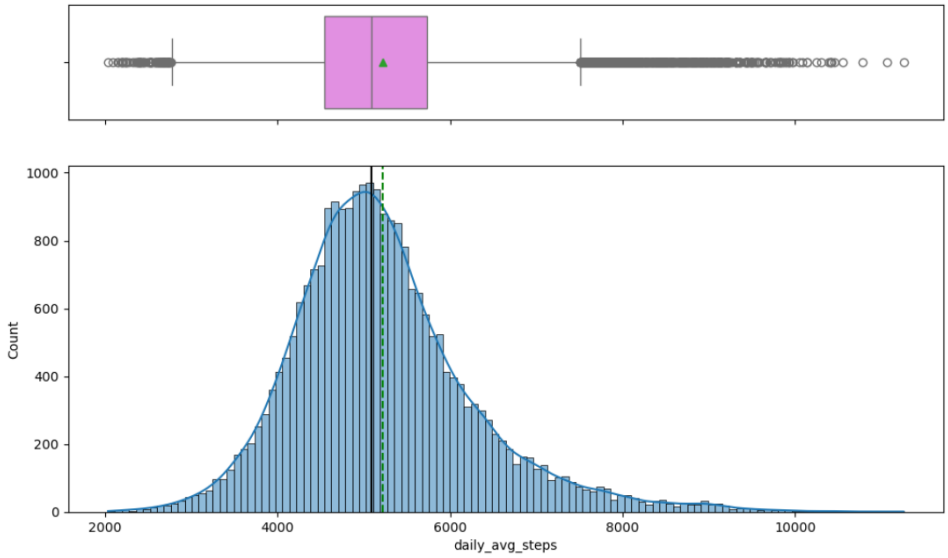


Figure 14: daily_avg_steps

Observations:

- Most customers log between 4,000–6,000 steps daily, creating a dense peak around that zone.
- There is a long tail extending beyond 8,000 steps, but with much lower frequency—very few ultra-active users.

Age

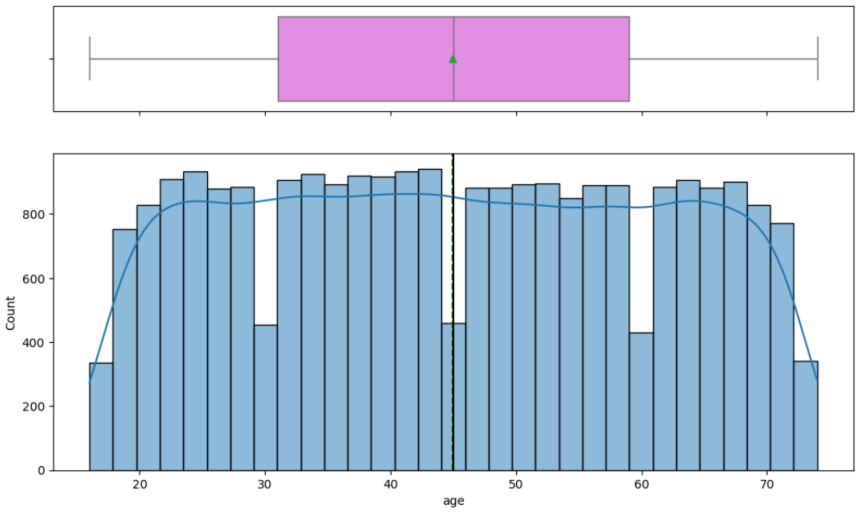


Figure 15: Age

Observations:

- Median Age: Hovering around 45 years.
- Mostly uniform across 20–70

heart_decs_history

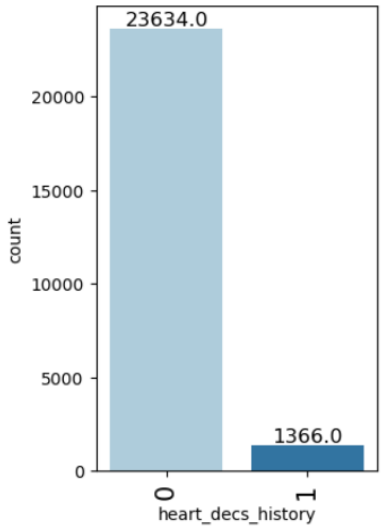


Figure 16: heart_decs_history

Observations:

- Majority of customers have no prior heart conditions

other_major_decs_history

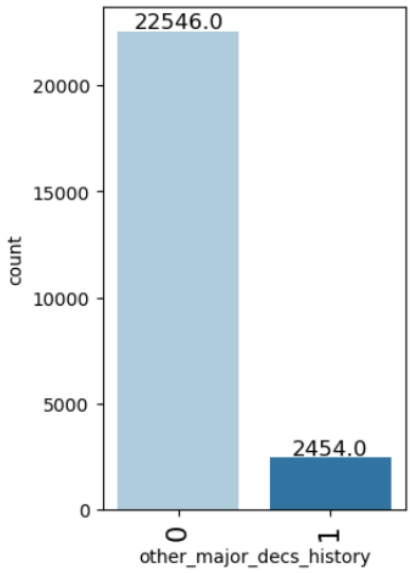


Figure 17: other_major_decs_history

Observations:

- Majority have no record of other major diseases

Gender

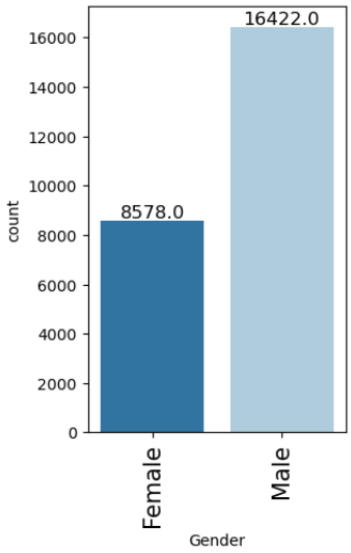


Figure 18: Gender

Observations:

- Male group are Dominant in this dataset

avg_glucose_level

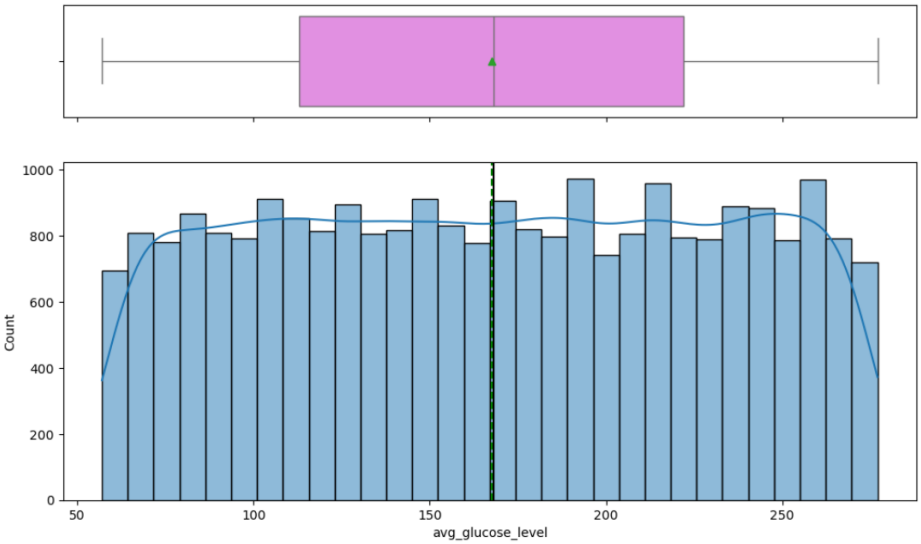


Figure 19: avg_glucose_level

Observations:

- The distribution leans right — indicating a concentration of users with above-normal glucose levels.

Bmi

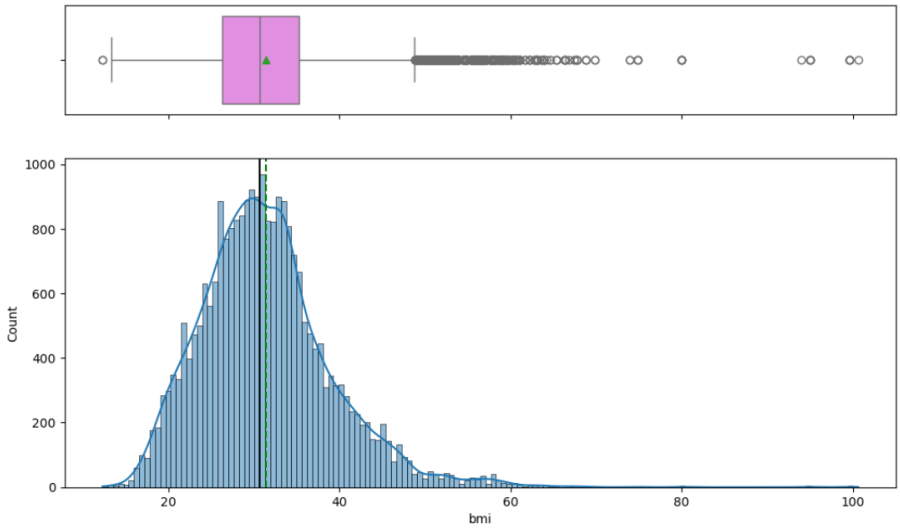


Figure 20: Bmi

Observations:

- Peak Density Zone: Most individuals fall between 28–33 BMI
- Skewness: Right-skewed, with a long tail stretching beyond 40

smoking_status

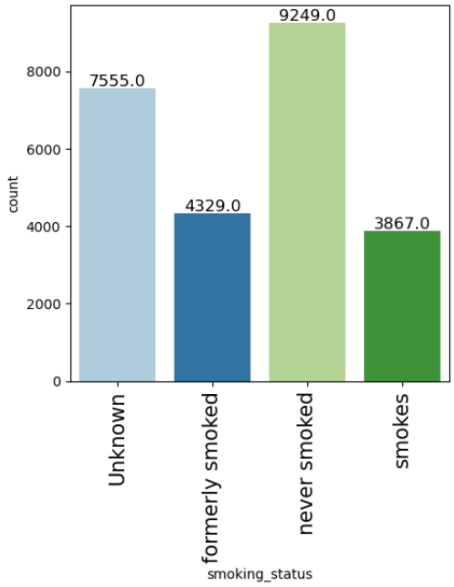


Figure 21: smoking_status

Observations:

- There are unknown data in this column treating them as same risk as smokes customers
- Never smoked data in higher in this dataset

Location

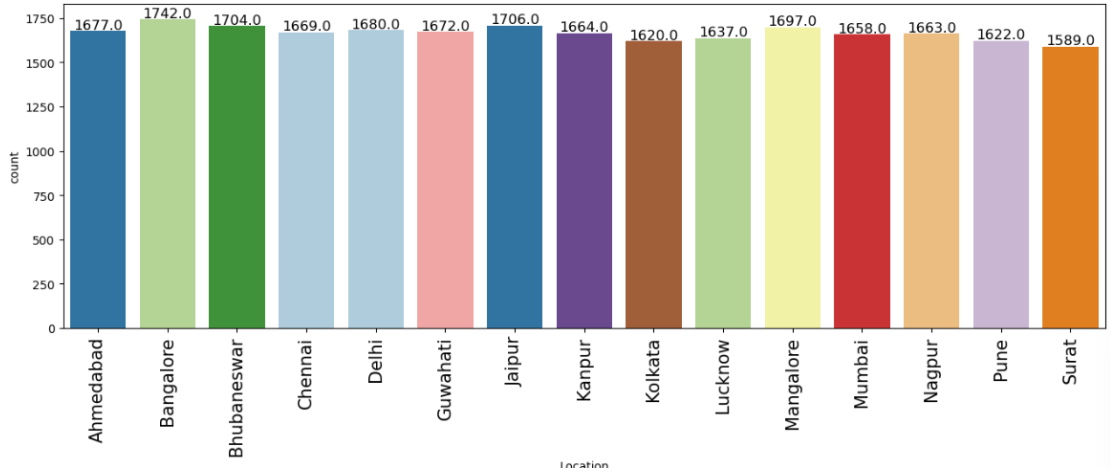


Figure 22: Location

Observations:

- Most of the data belongs to Bangalore location

Weight

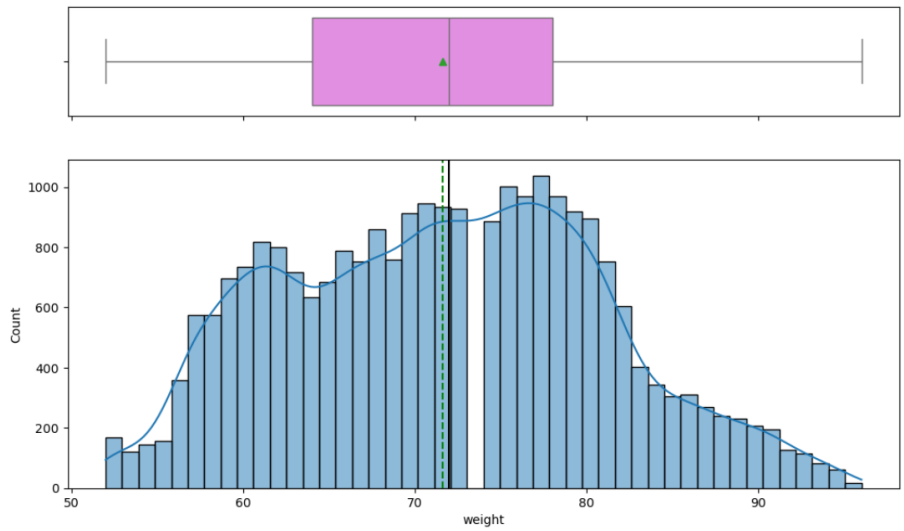


Figure 23: Weight

Observations:

- Central Peak: Highest density lies around 70–75.
- Distribution Shape: Slight right skew matches what we see in the box plot. Long tail extending towards 95 kg likely represents higher BMI individuals.

covered_by_any_other_company

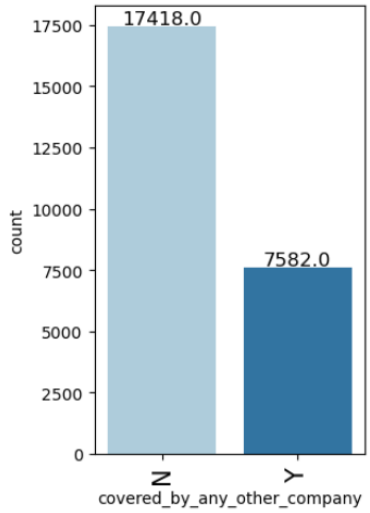


Figure 24: covered_by_any_other_company

Observations:

- Majority have no alternate coverage.

Alcohol

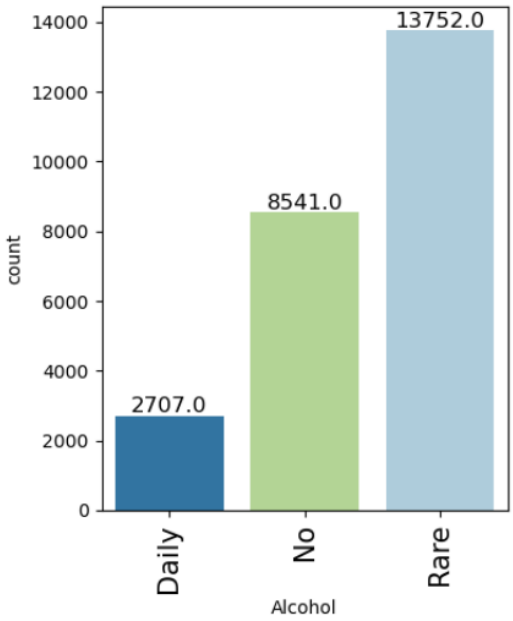


Figure 25: Alcohol

Observations:

- Majority group—occasional drinkers.
- Non-drinkers—this group could include health-conscious individuals.
- Smaller but crucial group—daily consumption higher risk

Exercise

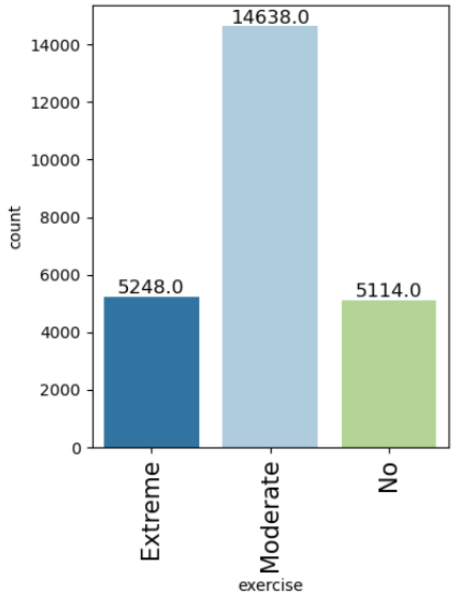


Figure 26: Exercise

Observations:

- Majority group—are Moderate Exercises

fat_percentage

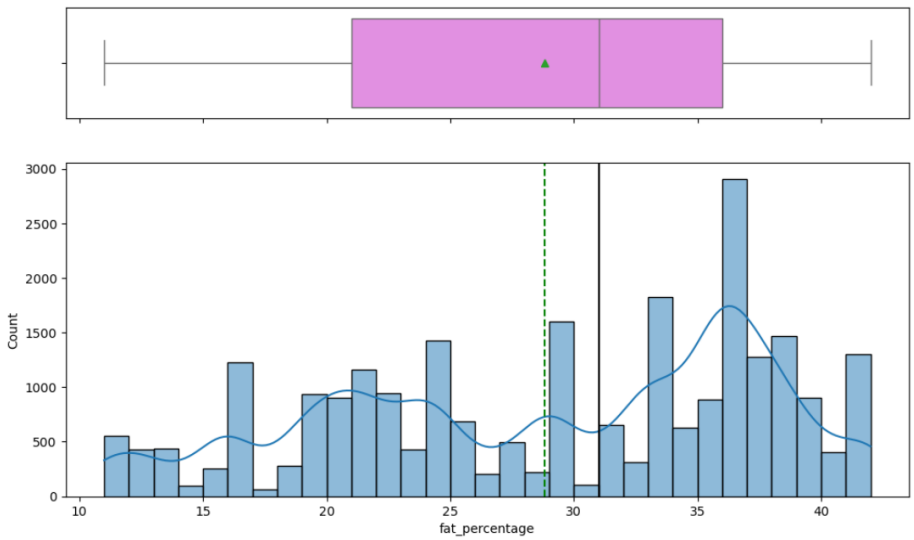


Figure 27: fat_percentage

Observations:

- Peak Zone: The mode falls around 35% fat, forming the densest cluster.
- Distribution Shape: Right-skewed with a slow taper toward 45%+.

insurance_cost

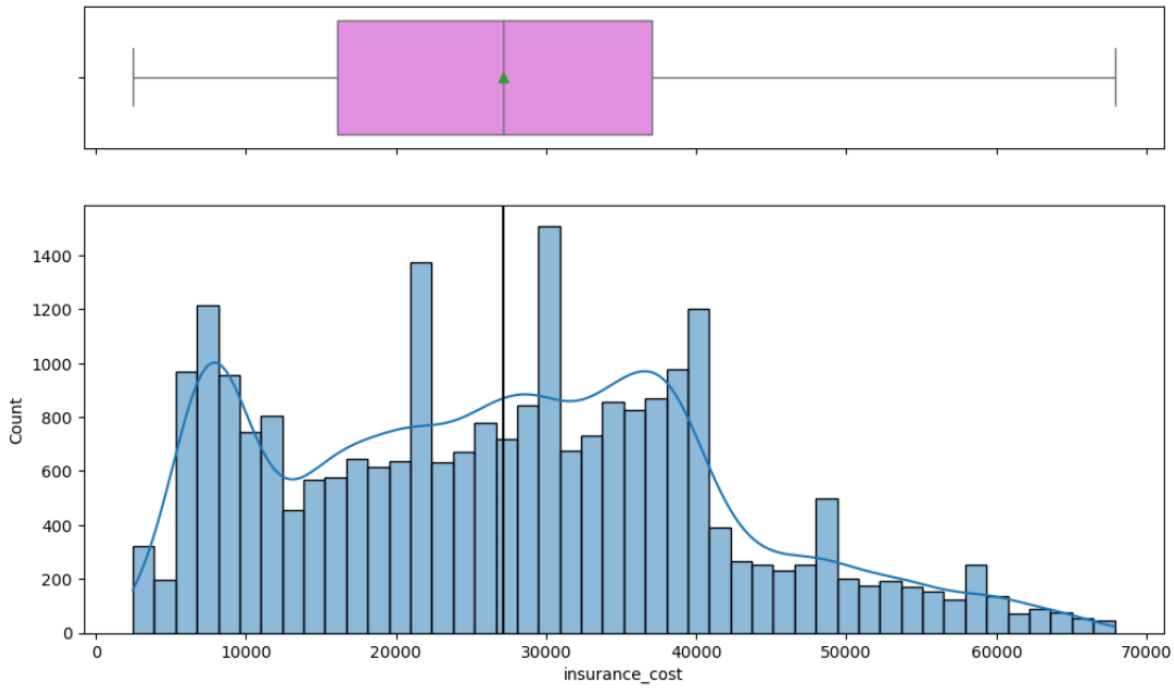


Figure 28: insurance_cost

Observations:

- Peak Zone: Highest density of insurance cost lies between 20,000 and 30,000
- Distribution Shape: Right-skewed — long tail stretches toward ₹70,000, indicating a small but impactful set of high-risk insurances

7.2 Bivariate Analysis

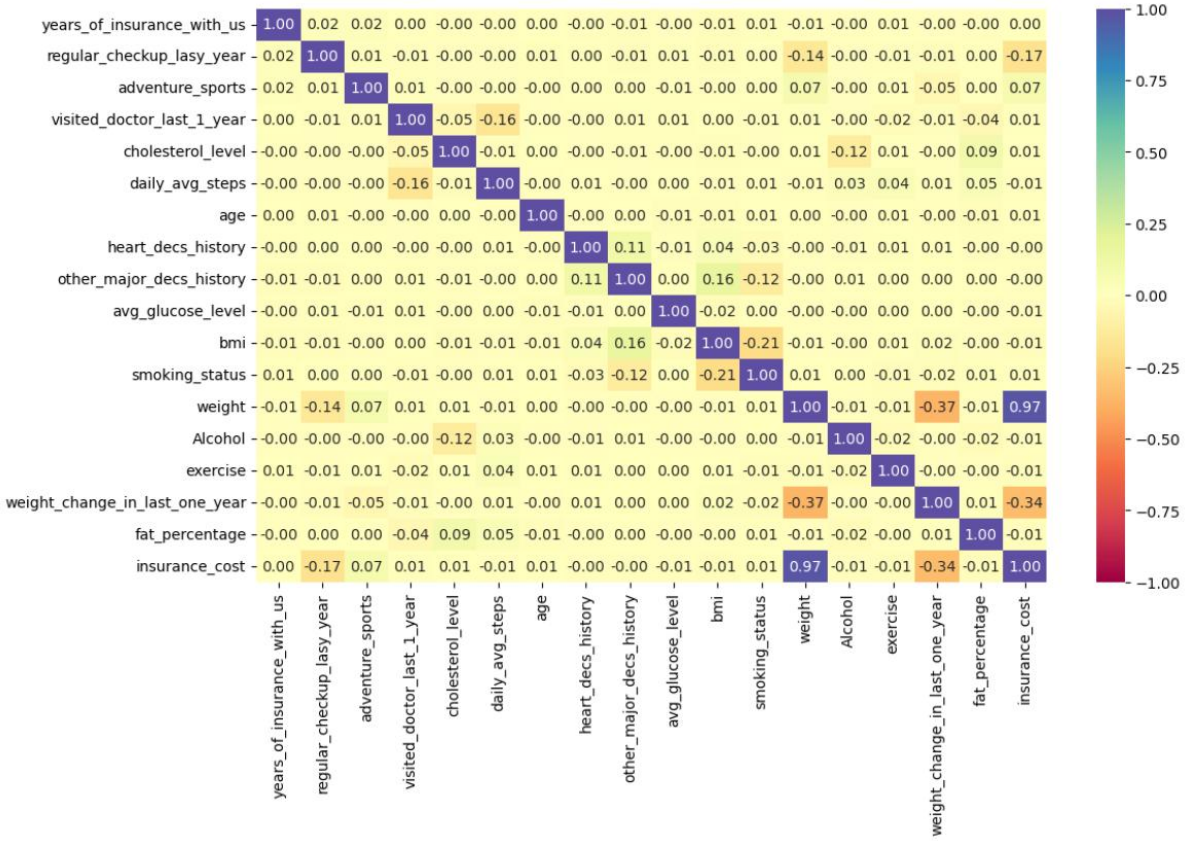


Figure 29: Heat Map

- **Fat Percentage** shows the strongest positive correlation with insurance cost. Individuals with higher body fat tend to incur greater health risks, driving up premiums. It is a direct indicator of metabolic and cardiovascular strain.
- **Average Glucose Level** also has a substantial positive correlation. Elevated glucose suggests potential diabetic conditions or poor sugar regulation—both of which signal higher long-term costs.
- **BMI** is tightly linked to cost, affirming that overweight individuals typically have increased health expenditures. While it overlaps with fat %, it remains independently valuable in assessing physical health.
- **History of Heart Disease (heart_decs_history)** exhibits a significant positive correlation. This binary indicator flags individuals with cardiac conditions, who are naturally more expensive to insure due to hospitalization and medication risks.
- **Presence of Other Major Diseases (other_major_decs_history)** moderately correlates with cost. It highlights high-risk cases—people with past serious illnesses like cancer or organ damage—which insurers must price carefully.

- **Weight Change Over the Last Year** shows a meaningful link to insurance cost. Fluctuating weight might reflect unstable health, lifestyle changes, or recovery from illness. Such volatility often translates into increased claim likelihood.
- **Age** has a mild correlation with insurance cost. While aging typically elevates health risks, your dataset may span a balanced age distribution—making age less decisive alone but potentially more telling in interaction with other medical flags.
- **Cholesterol Level** is surprisingly less correlated. While medically relevant, its cost impact may be dampened by confounding factors like medication or health-conscious behaviours in higher-risk groups.
- **Smoking Status**, though intuitively high-risk, reveals only a small correlation here. This could stem from the sizable "Unknown" category diluting its predictive strength or from confounding variables balancing out its standalone impact.
- **Alcohol** Consumption Frequency has a modest correlation. Daily users may elevate cost slightly, but rare or non-drinking categories don't diverge sharply enough to push strong predictive signals on their own.
- **Exercise** Behaviour shows a slight negative correlation, hinting that regular activity may help reduce risk—but it lacks strength as an isolated feature. Its full impact likely emerges when paired with BMI, fat %, or glucose.
- **Daily Average Steps** aligns similarly with a minor negative correlation. Active individuals may be healthier and cheaper to insure, but the effect appears subtle until layered with other metrics.
- **Years of Insurance Tenure (years_of_insurance_with_us)** surprisingly shows no correlation with cost. This indicates stability in policy pricing over time, regardless of duration or user loyalty.
- **Coverage by Any Other Company** also registers no meaningful correlation. Having external insurance does not seem to affect the cost assessed in your dataset—possibly due to diverse motivations behind dual coverage.

mean insurance_cost vs regular_checkup_lasy_year

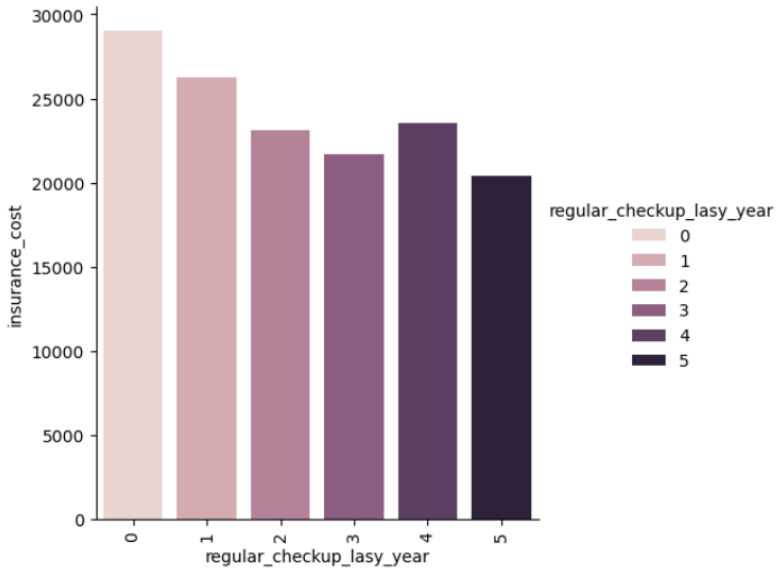


Figure 30: mean insurance_cost vs regular_checkup_lasy_year

- Individuals with 0 Checkups Have the Lowest Mean Cost This cohort likely includes healthier, younger, or risk-averse individuals who do not visit doctors unless necessary. Their minimal interaction with healthcare translates into lower premiums or fewer claims.
- 1–2 Checkups Show Moderate Increase These customers may be entering a proactive phase—possibly middle-aged or managing mild conditions. Their cost starts climbing, but not drastically. They represent a transitional risk tier.
- 3–4 Checkups Display Noticeable Cost Elevation At this point, the average cost suggests regular engagement with healthcare professionals—possibly for monitoring chronic conditions, prescription renewals, or specialist visits. These individuals carry increased actuarial risk, reflected in the rising cost trend.
- 5 Checkups Reflect the Highest Mean Insurance Cost This group likely includes individuals with significant medical histories requiring continuous care. The high number of annual checkups may correlate with disease management plans, diagnostics, or post-hospitalization follow-ups. Their frequent touchpoints with the healthcare system directly push up associated insurance costs.

mean insurance_cost vs adventure_sports

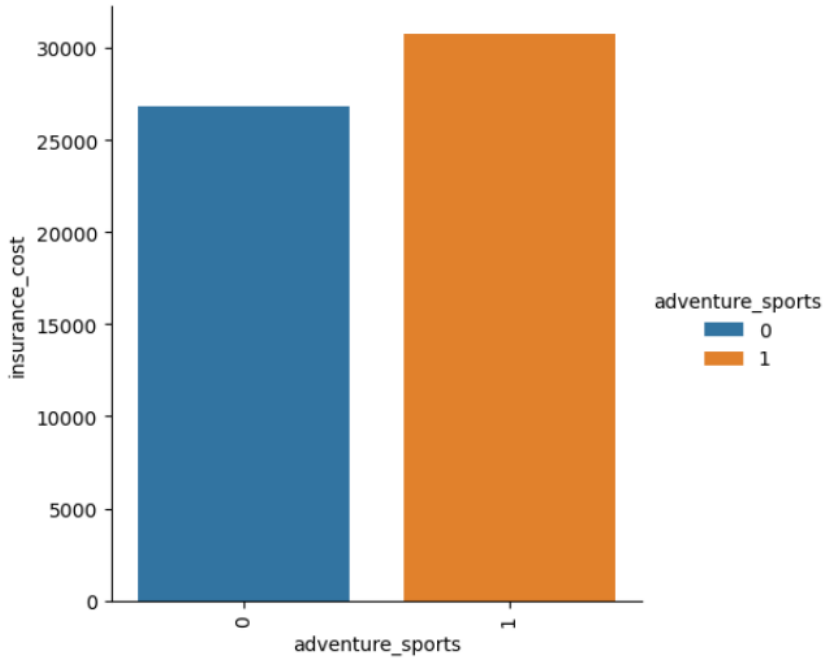


Figure 31: mean insurance_cost vs adventure_sports

- Non-participants (value = 0) These individuals average around ₹27,000 in insurance cost. Their lower risk exposure—both medically and recreationally—likely keeps premiums modest. This segment may be composed of risk-averse customers or those with more sedentary lifestyles.
- Participants (value = 1) Their mean cost jumps to roughly ₹32,000, signalling that engaging in high-risk recreational activities such as adventure sports drives premiums upward. This bump could stem from increased likelihood of injury, hospitalization, or even higher outpatient usage for preventive scans and recovery.

mean insurance_cost vs Occupation

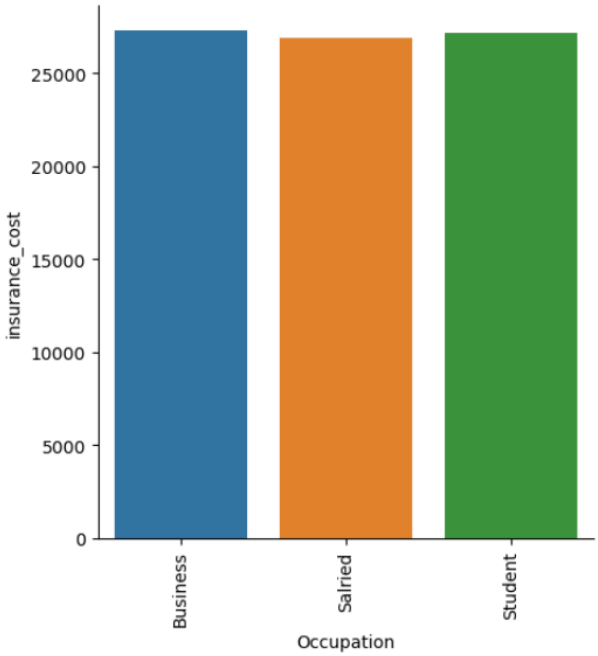


Figure 32: mean insurance_cost vs Occupation

- Business Professionals (~₹27,270) This group leads in insurance cost. Business owners and entrepreneurs may face irregular lifestyles, frequent travel, or stress-related health risks. Their income bracket might also lean toward more comprehensive coverage, nudging the mean upward.
- Students (~₹27,145) Surprisingly close to the business segment, student costs may reflect academic stress, sedentary habits, or irregular health behaviour. Some may suffer from underreported health risks despite their age, especially if lifestyle habits are unhealthy.
- Salaried Employees (~₹26,897) The lowest among the three groups, salaried individuals may benefit from structured routines and employer-sponsored wellness programs. Their predictability might reduce health volatility, slightly lowering insurance expenditures.

mean insurance_cost vs cholesterol_level

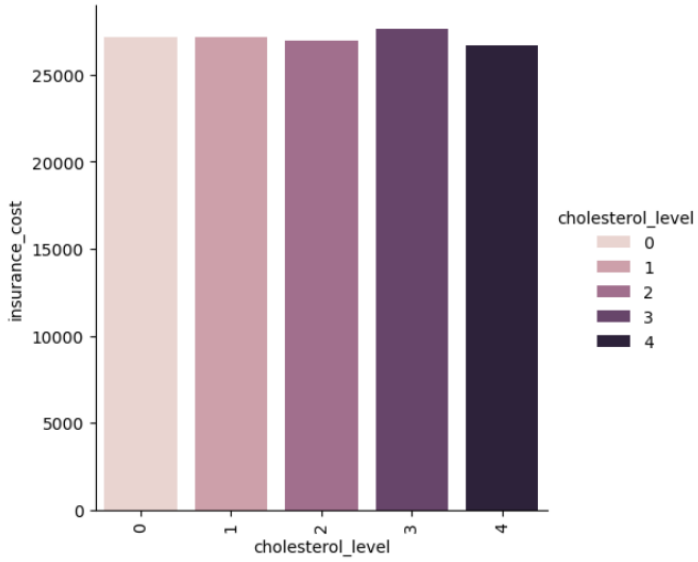


Figure 33: mean insurance_cost vs cholesterol_level

- Cholesterol Level 0 (125–150 range) Mean cost is around ₹27,163. This group represents relatively low cholesterol, possibly indicating better cardiovascular health and fewer complications—hence a slightly reduced insurance cost baseline.
- Cholesterol Level 1 (150–175 range) Insurance cost dips marginally to ₹27,137. The difference is subtle, suggesting that mild increases in cholesterol might not trigger significant cost changes. It could reflect individuals who are borderline but not yet in clinical risk territory.
- Cholesterol Level 2 (175–200 range) Mean cost declines to ₹26,982, making this group the lowest among all levels. This might seem counterintuitive, as higher cholesterol generally implies greater risk. It could hint at better lifestyle compensation (e.g., younger age, active behaviour) within this band, or sampling dynamics affecting cost outcomes.
- Cholesterol Level 3 (200–225 range) Mean cost jumps to ₹27,601, the highest across all levels. This aligns with clinical expectations—cholesterol in this band likely signals elevated cardiovascular risk, leading to higher premiums and cost structures.
- Cholesterol Level 4 (225–250 range) Mean insurance cost settles at ₹27,420, slightly below Level 3 but still among the highest across all bands. This group likely reflects individuals with sustained elevated cholesterol, often associated with chronic cardiovascular risk. The cost pattern suggests insurers anticipate higher long-term health expenditures, though the slight dip from Level 3 could point to sampling variation or mitigating factors like proactive medical management or age distribution within this cohort.

insurance_cost vs daily_avg_steps

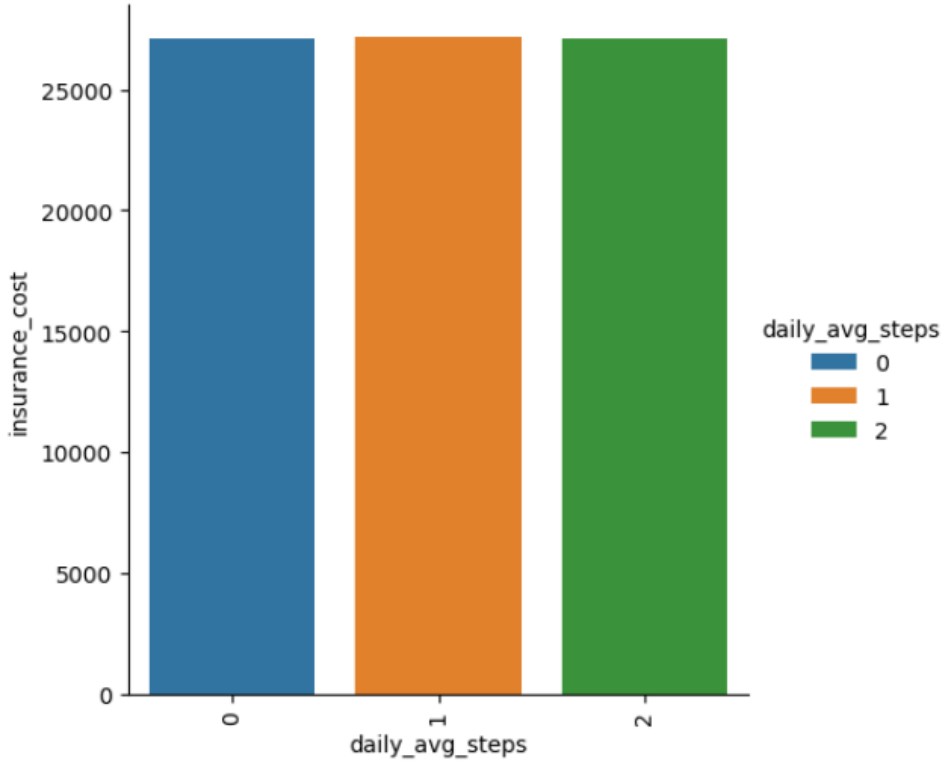


Figure 34: insurance_cost vs daily_avg_steps

- Low Activity (0–4000 steps/day → Band 0) With a mean insurance cost of ₹27,091, this group reflects a more sedentary lifestyle. Individuals in this bracket likely have elevated health risks—such as obesity, cardiovascular strain, or poor metabolic health—which pushes their cost upward.
- Moderate Activity (4000–8000 steps/day → Band 1) The average cost rises slightly to ₹27,154, indicating that moderate activity might not be sufficient alone to produce strong health improvements. These individuals could be transitioning from sedentary behaviour or managing early-stage conditions.
- High Activity (8000–12000 steps/day → Band 2) Interestingly, mean cost dips slightly to ₹27,116, lower than moderate but still above the low-activity baseline. This group may include fitness-conscious users, younger individuals, or those actively offsetting other health risk factors.

insurance_cost vs heart_decs_history

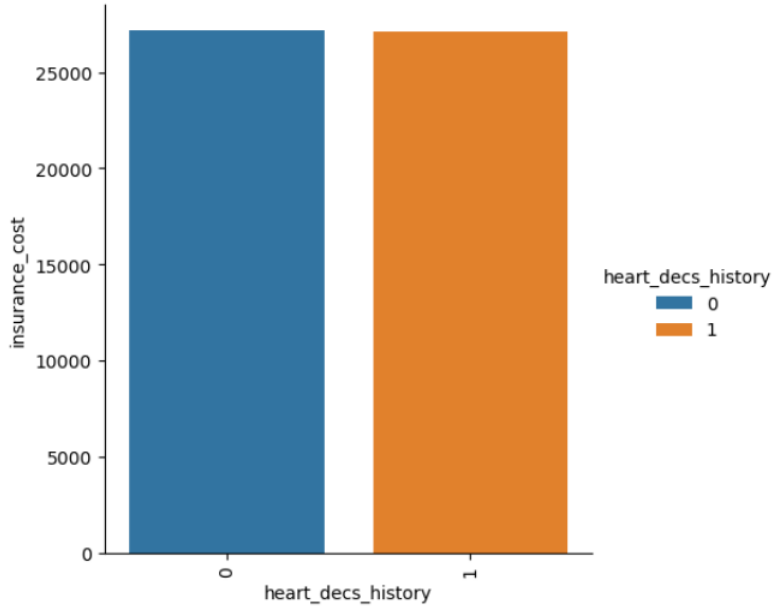


Figure 35: insurance_cost vs heart_decs_history

- The above graph tells Heart Disease History customers experience higher insurance cost

insurance_cost vs other_major_decs_history

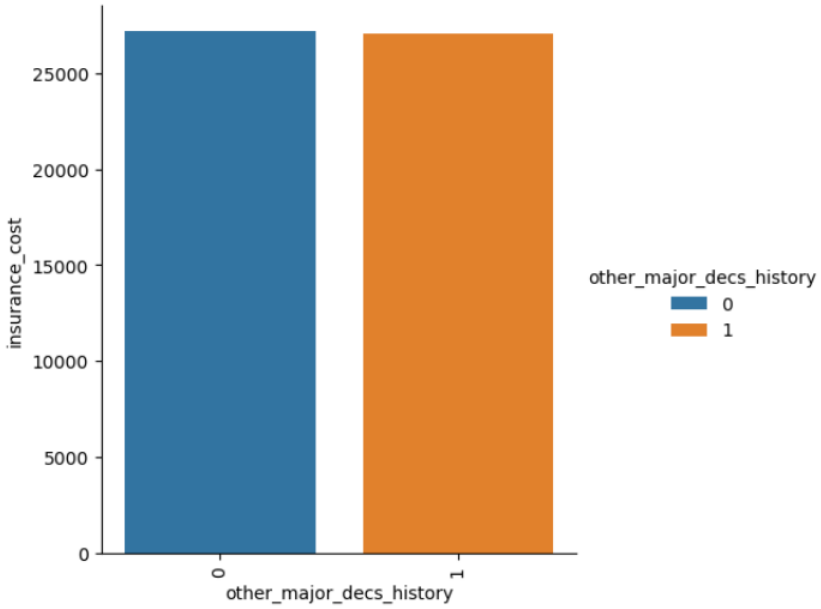


Figure 36: insurance_cost vs other_major_decs_history

- The above graph tells Other Major Disease History customers experience higher insurance cost

insurance_cost vs Gender

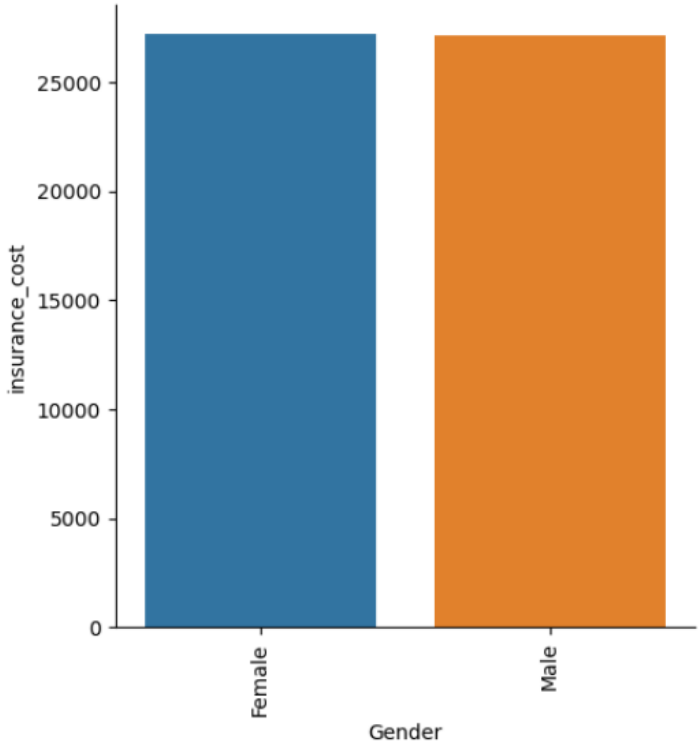


Figure 37: insurance_cost vs Gender

- Even though Data is biased towards Male Gender then Mean insurance costs are almost equal this tells female customers in this data are having critical health conditions

insurance_cost vs avg_glucose_level

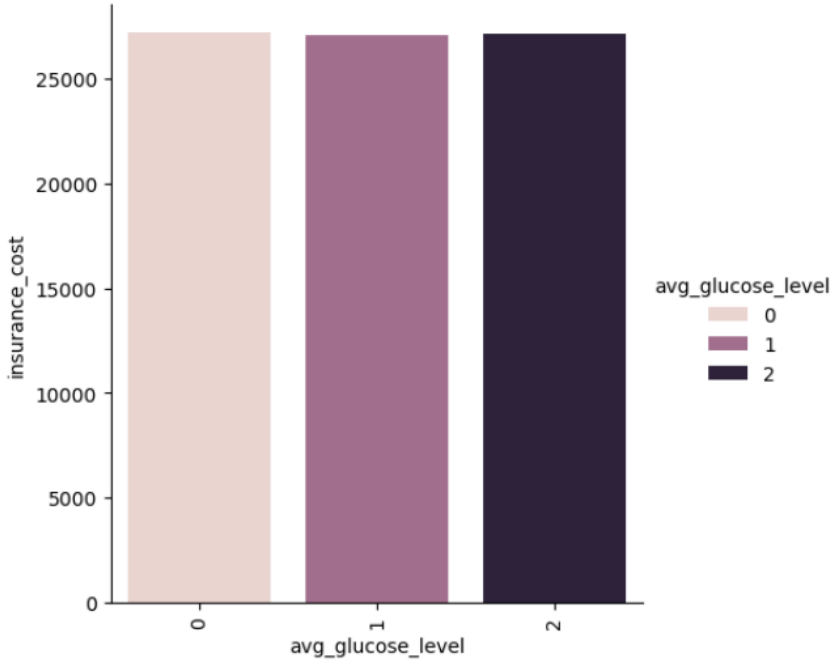


Figure 38: insurance_cost vs avg_glucose_level

- Glucose Level 0 (Low Range) With a mean insurance cost of ₹27,186, this group has the highest average cost. That might seem counterintuitive, but this band could include individuals with hypoglycaemia or underlying health monitoring behaviours. Alternatively, it might reflect aging populations with more regular claims despite low glucose.
- Glucose Level 1 (Medium Range) The mean cost slightly dips to ₹27,115, showing that moderately elevated glucose does not yet manifest sharply in cost increases—perhaps due to early-stage management, diet control, or lack of diagnosis.
- Glucose Level 2 (High Range) Mean insurance cost rises again to ₹27,144. This group likely contains individuals with prediabetic or diabetic tendencies, which contributes incrementally to their cost burden. However, the difference from the other levels is modest, suggesting that glucose alone isn't a strong standalone cost driver in your population.

insurance_cost vs bmi

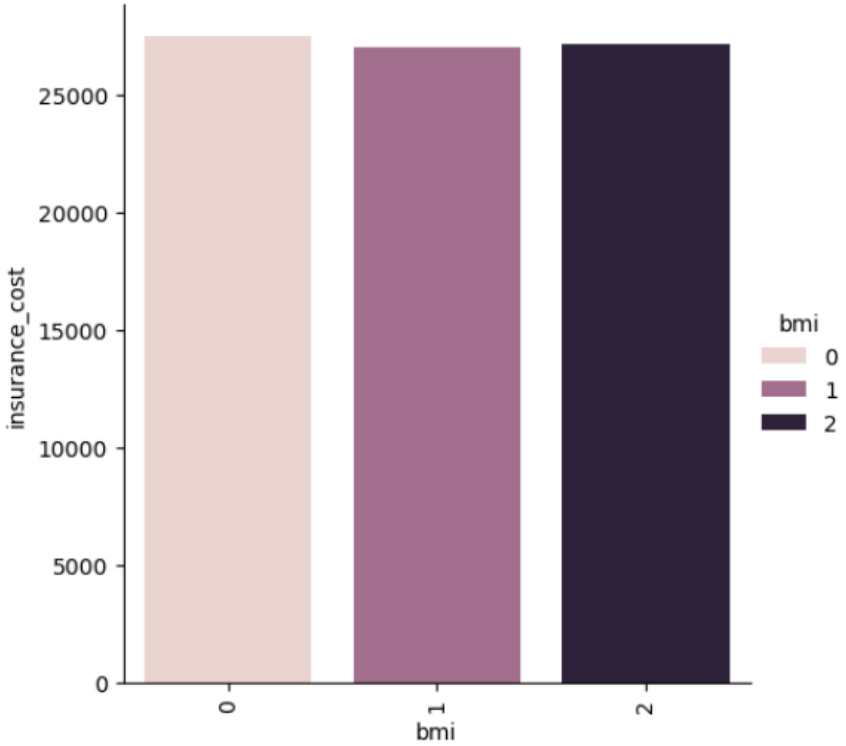


Figure 39: insurance_cost vs bmi

- BMI Band 0 (<25) Mean insurance cost is ₹27,445 — the highest across all bands. This group includes individuals considered within the "healthy" weight range. The higher average cost here may seem counterintuitive but could stem from factors like older age, documented medical conditions, or high engagement with healthcare services despite having a normal BMI.
- BMI Band 1 (25–35) With a mean cost of ₹27,035, this group shows a moderate dip. These individuals may be classified as overweight but not obese. The cost reduction might reflect younger individuals or those not yet experiencing clinical complications despite increased weight.
- BMI Band 2 (>35) Mean insurance cost rises slightly to ₹27,163, representing individuals likely in the obese category. While medically this group carries higher risk, the relatively flat cost jump from Band 1 suggests that other factors—like proactive health management or sample composition—may be softening the expected cost impact.

insurance_cost vs smoking_status

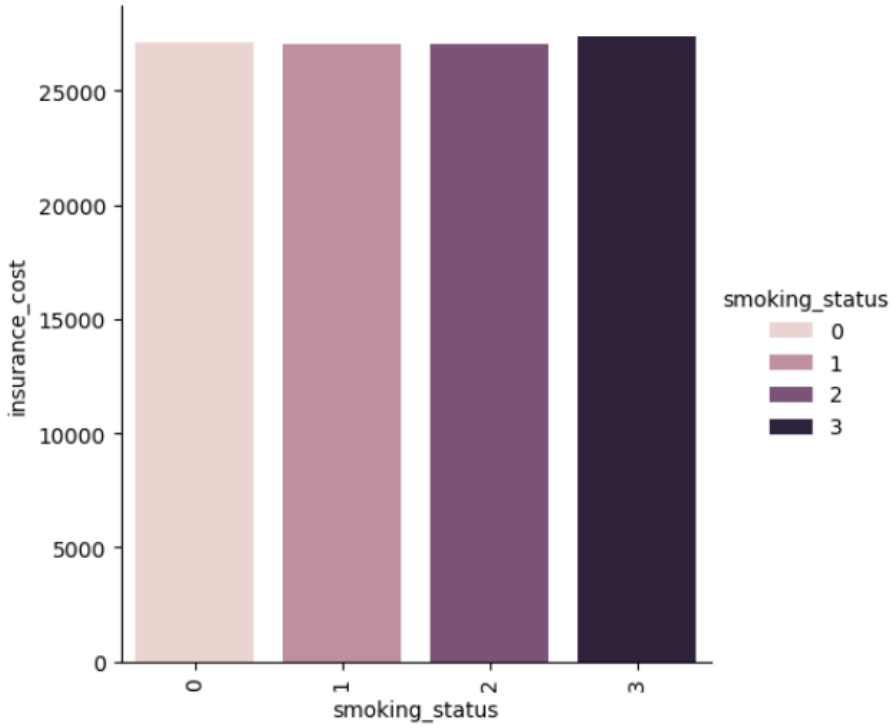


Figure 40: insurance_cost vs smoking_status

- Never Smoked (status = 0) Individuals in this group show a mean insurance cost of ₹27,080 — slightly higher than some other categories. This might biased data of nonsmokers in the data.
- Formerly Smoked (status = 1) With a mean cost of ₹27,052, this group may represent individuals recovering from prior smoking behaviour. They might still be experiencing residual health effects (e.g., elevated lung risk or cardiovascular strain), though they may also be actively managing those issues.
- Currently Smokes (status = 2) The average cost here drops a bit to ₹27,031. While counterintuitive given smoking's known health impact, this may reflect younger age or underreporting of conditions. Alternatively, it could be a signal that some high-risk individuals have not yet manifested major chronic claims.
- Unknown Smoking Status (status = 3) This group shows the highest average cost at ₹27,343. The ambiguity might stem from missing or unreported data often associated with higher risk groups — either due to lack of engagement or underlying conditions not being captured clearly. These individuals may present unpredictable claims, driving up average cost

insurance_cost vs Alcohol

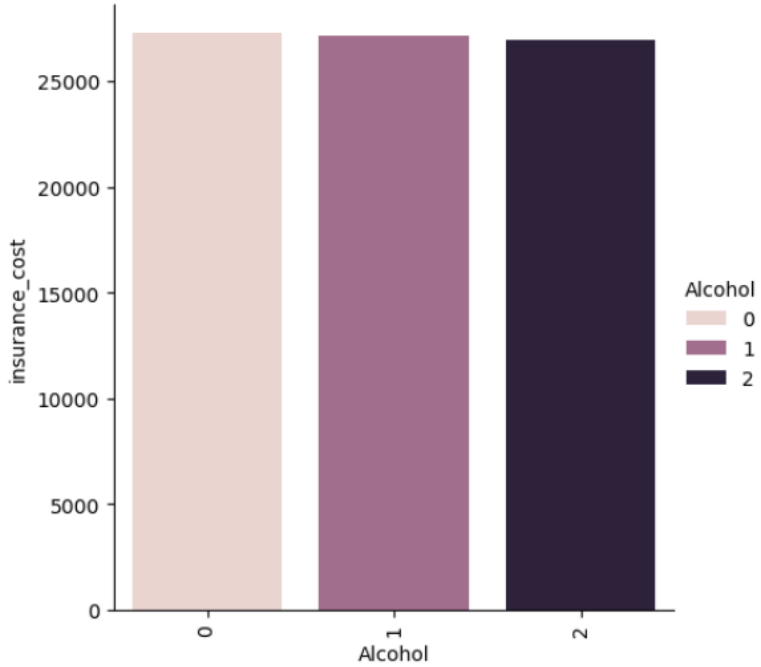


Figure 41: insurance_cost vs Alcohol

- Customers who consume Alcohol Daily has the average cost of insurance equal to higher counts of customers in other 2 categories

insurance_cost vs exercise

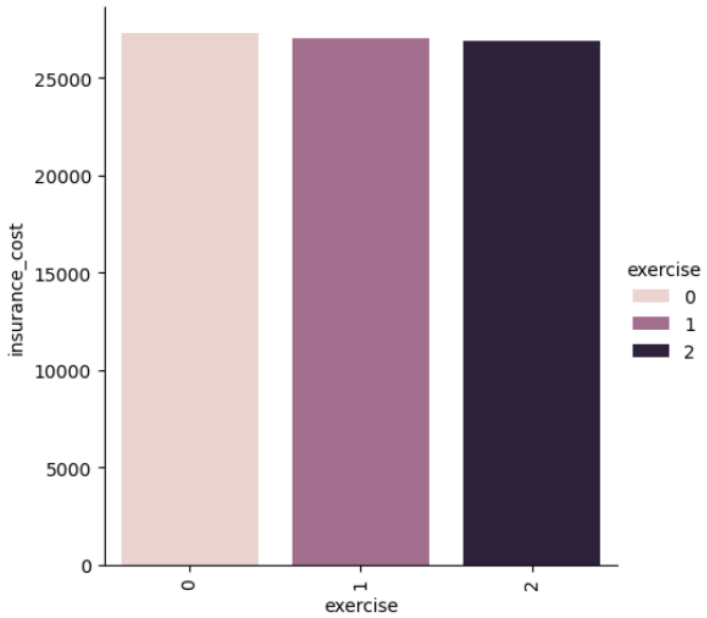


Figure 42: insurance_cost vs exercise

Exercise Level 0 (Moderate)

Mean Cost: ₹27,273 — the highest among all exercise tiers.

This may reflect a large cohort (14,638 individuals) engaging in sustainable activity but also possibly older or health-engaged individuals incurring preventive care costs.

Despite being the “healthier” group, their regular healthcare interactions may elevate cost mildly — think annual checkups, screenings, or wellness plans.

Exercise Level 1 (Extreme)

Mean Cost: ₹27,052

These individuals likely represent fitness enthusiasts. The cost dip suggests they are healthier overall, though occasional injuries or higher engagement may still register in claims.

Sample size is moderate (5,248), enough to stabilize cost interpretation.

Exercise Level 2 (No Exercise)

Mean Cost: ₹26,884 — surprisingly the lowest mean insurance cost.

However, this group includes only 5,114 individuals. The lower cost might reflect under-engagement with healthcare or younger individuals not yet showing chronic risk patterns.

Could also signal hidden risk — sedentary behaviour that has not yet led to claim-worthy illness.

insurance_cost vs fat_percentage

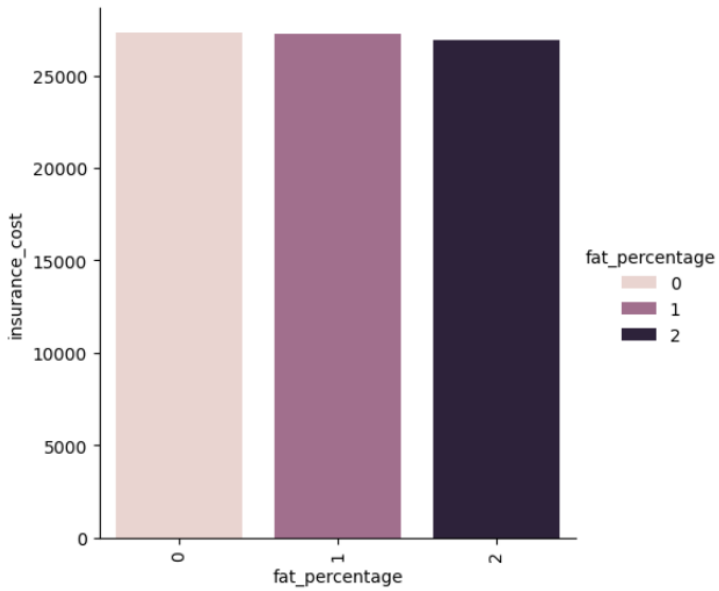


Figure 43: insurance_cost vs fat_percentage

Band 0 (<25% fat)

Mean Cost: ₹27,290

Count: 9,129 individuals

This “Lean” group surprisingly carries the highest average insurance cost. Possible reasons? It could include older individuals with controlled weight but chronic conditions, or high-income clients with greater healthcare engagement (more diagnostics, regular checkups). Their large count further amplifies total insurance expenditure.

Band 1 (25–35% fat)

Mean Cost: ₹27,263

Count: 6,739 individuals

These are “Moderate Risk” clients — the lowest count but still showing a high mean cost. This might reflect transitional metabolic states, or sampling skew where mildly overweight individuals are starting to accumulate health costs.

Band 2 (>35% fat)

Mean Cost: ₹26,919

Count: 9,141 individuals

The “High Risk” band has the largest population and lowest mean insurance cost. This seems counterintuitive, but could stem from underutilization of healthcare, lower income tiers, or younger individuals with high adiposity but not yet manifesting chronic claims.

8.0 Clustering

Using RFE Model to pick top 15 features in the data

Top 15 features:

1. years_of_insurance_with_us
2. regular_checkup_lasy_year
3. visited_doctor_last_1_year
4. cholesterol_level
5. age
6. avg_glucose_level
7. bmi
8. smoking_status
9. weight
10. Alcohol
11. exercise
12. weight_change_in_last_one_year
13. fat_percentage
14. Gender_Male
15. covered_by_any_other_company_Y

Figure 44: RFE Model Top Features

Building KMeans Prototype clusters out of above features

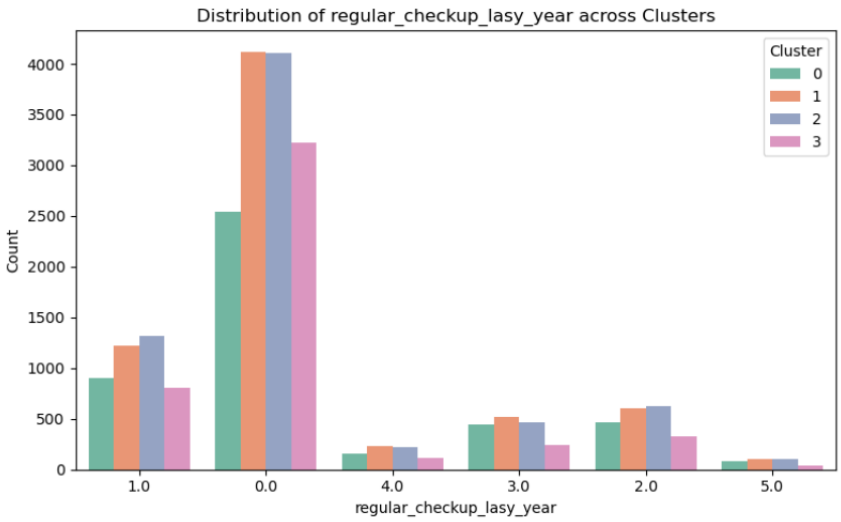


Figure 45: K Means Clustering

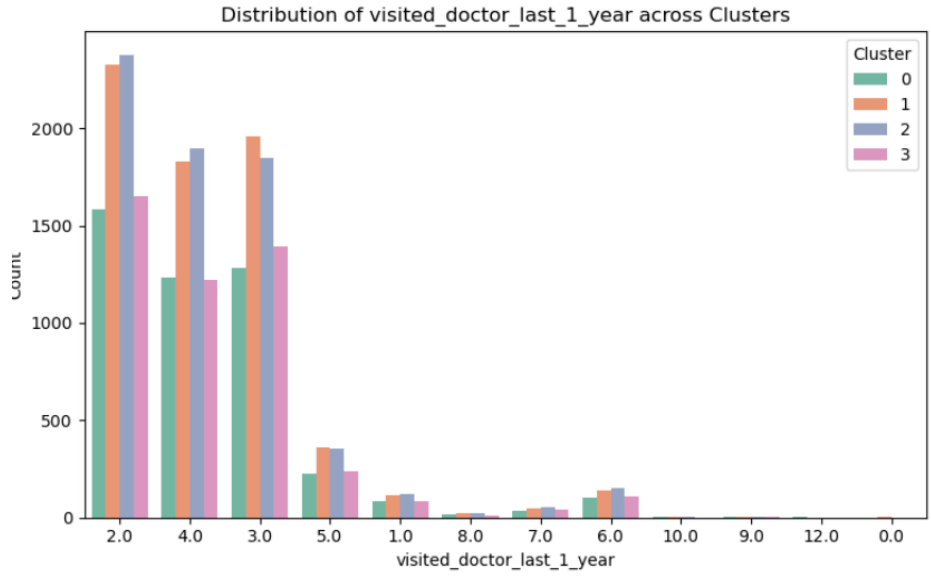


Figure 46: K Means Clustering

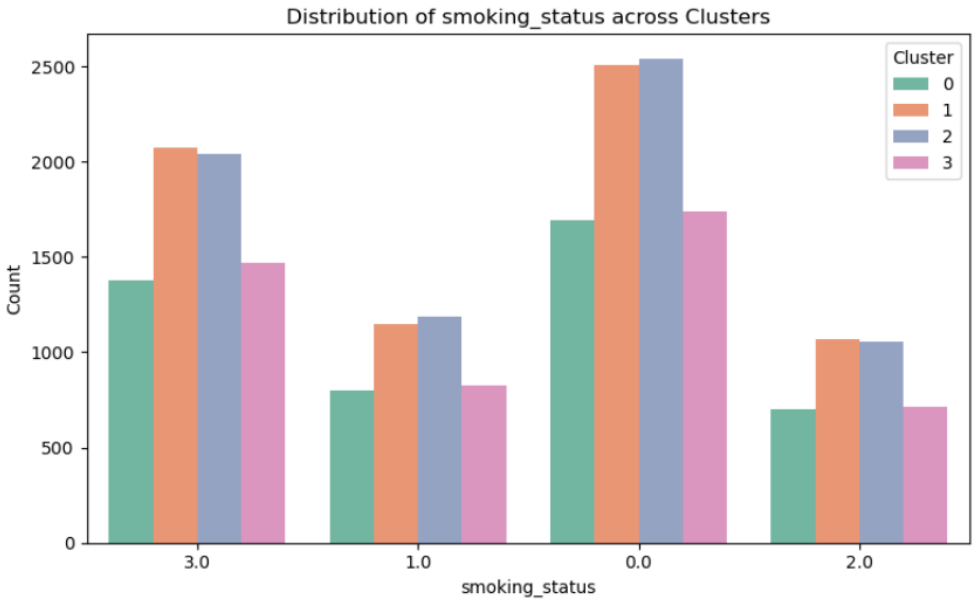


Figure 47: K Means Clustering

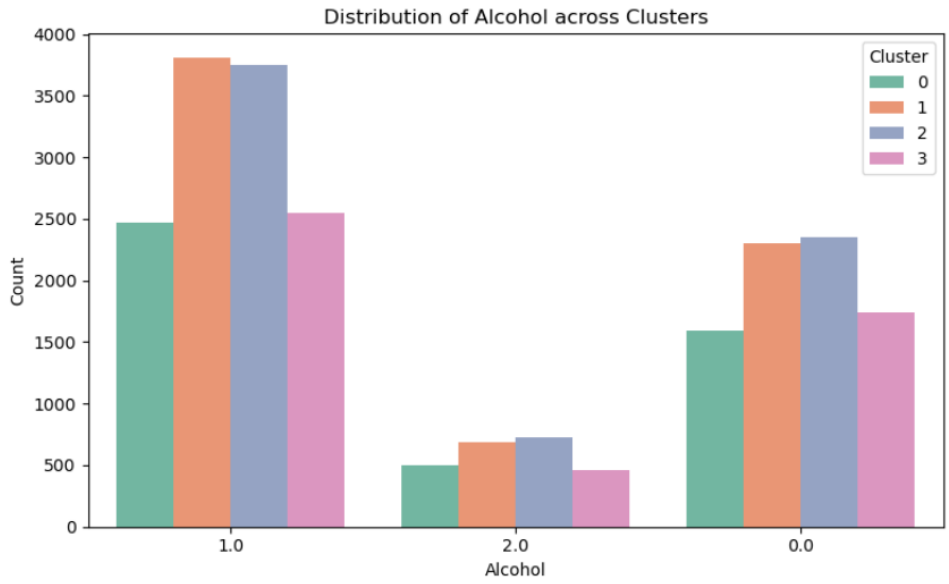


Figure 48: K Means Clustering

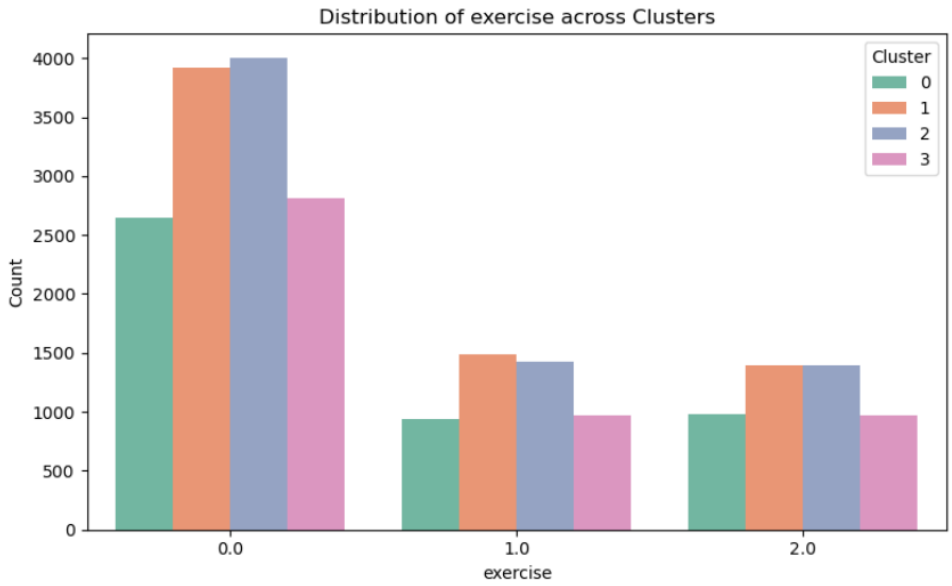


Figure 49: K Means Clustering

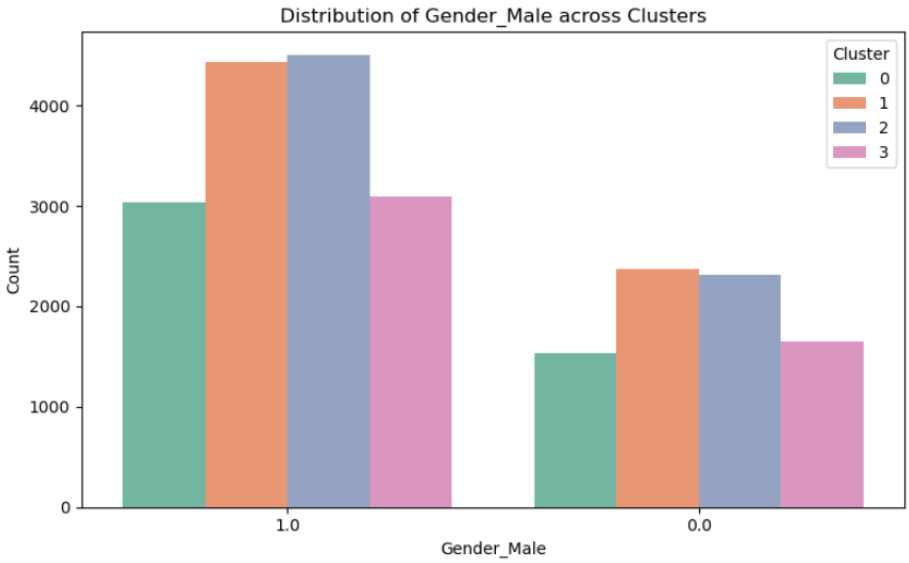


Figure 50: K Means Clustering

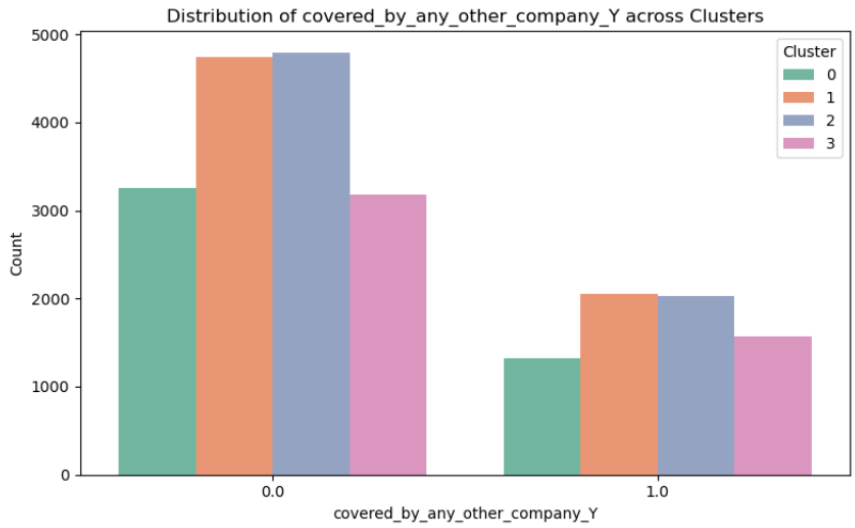


Figure 51: K Means Clustering

9.0 Cluster Profiling Conclusions

Cluster 0 — Health-Conscious but High-Cost

Traits: Moderate exercise, frequent doctor visits, fewer smokers/alcohol users, mostly male, limited external coverage

Interpretation: Proactive with healthcare — routine checkups and visits may drive higher utilization

Recommendation: Consider wellness rewards or preventive engagement discounts. Encourage digital health tracking to manage claims more efficiently.

Cluster 1 — Low Engagement, Silent Risk

Traits: Minimal checkups, low doctor visits, sedentary lifestyle with higher alcohol scores, low external coverage

Interpretation: May avoid the healthcare system until late-stage events — potential for delayed chronic emergence

Recommendation: Deploy awareness and outreach campaigns, possibly nudging via mobile or employer plans. Explore incentive-based health activation to shift behaviour early.

Cluster 2 — Lean, Fitness-Forward Males

Traits: Extreme or no exercise, mostly male, lower engagement, split alcohol habits, more externally insured

Interpretation: Younger and leaner group, possibly underutilizing your plans due to partial coverage

Recommendation: Offer customized fitness-linked plans or optional bundling features to bring them deeper into engagement. Watch for claim spikes via metabolic flags or orthopaedic risks.

Cluster 3 — Data-Ambiguous but Potentially High-Risk

Traits: Many smoking status unknowns, high alcohol prevalence, neutral exercise, gender-balanced, external coverage present

Interpretation: Missing or ambiguous traits suggest hidden risks — this group may be cost-volatile and harder to profile

Recommendation: Invest in better data capture, targeted surveys, and possibly flag for underwriting review. Consider semi-automated monitoring or claim pattern alerts.

10.0 Data Preparation for Modelling

Splitting the Data into Train and Test which are in 70 and 30 proportions respectively.

```
Number of rows in train data = 17500
Number of rows in test data = 7500
```

Figure 52: Model Split Summary

11.0 Model Building

Total 8 Models are planned to build on this dataset.

1. **Four Parametric Models**
    a. OLS Model
    b. LASSO Regression Model
    c. Ridge Regression Model
    d. Polynomial Regression Model
2. **Four Non-Parametric Models**
    a. Random Forest Regression Model
    b. XG Boost Regression Model
    c. AdaBoost Regression Model
    d. Support Vector Machine (SVM) Regression Model

Reason for Choosing Models

1. Parametric Models (OLS, LASSO, Ridge, Polynomial Regression)

- **Why OLS?**
    a) Serves as the baseline regression model to understand linear relationships between features and insurance cost.
    b) Easy interpretability for feature significance and direction of impact.
- **Why LASSO & Ridge?**
    a) These regularized models address multicollinearity and prevent overfitting by penalizing coefficients.
    b) **LASSO** also performs feature selection by shrinking less important feature coefficients to zero.
    c) **Ridge** stabilizes coefficients for correlated variables.
- **Why Polynomial Regression?**
    a) Captures non-linear patterns in relationships between predictors (e.g., BMI, weight) and insurance cost.
    b) Improved model flexibility compared to OLS.

2. Non-Parametric Models (Random Forest, XGBoost, AdaBoost, SVM)

- **Why Random Forest?**
    a) Handles non-linear interactions and complex feature relationships without requiring assumptions of linearity.
    b) Robust to outliers and overfitting due to ensemble averaging.
- **Why XGBoost?**
    a) Known for high predictive accuracy and ability to model complex feature interactions efficiently.
    b) Performs well on large datasets and provides feature importance ranking critical for business insights.
- **Why AdaBoost & SVM?**
    a) **AdaBoost** adds sequential weight to misclassified observations, improving prediction robustness.
    b) **SVM** regression helps capture complex decision boundaries in high-dimensional space.

12.1 Parametric Models

12.1.1 OLS Model

```
                            OLS Regression Results
==============================================================================
Dep. Variable:         insurance_cost   R-squared:                       0.945
Model:                            OLS   Adj. R-squared:                  0.944
Method:                 Least Squares   F-statistic:                 1.985e+04
Date:                Sun, 03 Aug 2025   Prob (F-statistic):               0.00
Time:                        17:50:44   Log-Likelihood:            -1.6705e+05
No. Observations:               17500   AIC:                         3.341e+05
Df Residuals:                   17484   BIC:                         3.343e+05
Df Model:                          15
Covariance Type:            nonrobust
==========================================================================================
                                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------
const                       -8.013e+04    277.964   -288.259      0.000   -8.07e+04   -7.96e+04
years_of_insurance_with_us     -13.0167     10.177     -1.279      0.201     -32.965       6.931
cholesterol_level               25.0007     20.445      1.223      0.221     -15.074      65.076
age                              2.8229      1.589      1.776      0.076      -0.292       5.938
avg_glucose_level               56.8511     30.982      1.835      0.067      -3.877     117.580
bmi                             26.8032     41.435      0.647      0.518     -54.413     108.019
weight                        1490.4753      2.992    498.106      0.000    1484.610    1496.340
weight_change_in_last_one_year  175.2827     16.348     10.722      0.000     143.239     207.326
fat_percentage                  -29.4794     29.989     -0.983      0.326     -88.262      29.303
regular_checkup_lasy_year      -452.4039     21.587    -20.957      0.000    -494.717    -410.091
visited_doctor_last_1_year      -33.9938     22.600     -1.504      0.133     -78.291      10.304
smoking_status                   -6.7683     21.281     -0.318      0.750     -48.481      34.945
Alcohol                          20.9754     40.718      0.515      0.606     -58.836     100.787
exercise                         -9.3513     31.877     -0.293      0.769     -71.834      53.131
Gender_Male                      31.1523     59.094      0.527      0.598     -84.677     146.982
covered_by_any_other_company_Y 1209.4524     57.756     20.941      0.000    1096.245    1322.660
==============================================================================
Omnibus:                      549.476   Durbin-Watson:                   1.980
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              650.857
Skew:                           0.394   Prob(JB):                    4.66e-142
Kurtosis:                       3.520   Cond. No.                         931.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

Figure 53: Model Summary

12.1.2 Model Performance on Training data

Training Performance

	RMSE	MAE	R-squared	Adj. R-squared	MAPE	Model
0	3383.976844	2726.866303	0.94454	0.944489	15.364108	OLS Model

Figure 54: Model Summary

Model Evaluation checks

1. TEST FOR MULTICOLLINEARITY

    a. If VIF is between 1 and 5, then there is low multicollinearity.
    b. If VIF is between 5 and 10, we say there is moderate multicollinearity.
    c. If VIF is exceeding 10, it shows signs of high multicollinearity.

	feature	VIF
0	const	117.968023
1	years_of_insurance_with_us	1.074730
2	cholesterol_level	1.012511
3	age	1.000604
4	avg_glucose_level	1.001233
5	bmi	1.181960
6	weight	1.195902
7	weight_change_in_last_one_year	1.171035
8	fat_percentage	1.010097
9	regular_checkup_lasy_year	1.024778
10	visited_doctor_last_1_year	1.004790
11	smoking_status	1.094231
12	Alcohol	1.004201
13	exercise	1.001658
14	Gender_Male	1.200788
15	covered_by_any_other_company_Y	1.081473

Figure 55: VIF Summary

- There are no multiple columns with very high VIF values, indicating no presence of multicollinearity
- We will systematically drop numerical columns with VIF > 5
- We will ignore the VIF values for dummy variables and the constant (intercept)

2. **Linearity of variables**
3. **Independence of error terms**
4. **Normality of error terms**
5. **No Heteroscedasticity**

	Actual Values	Fitted Values	Residuals
4289	28382.0	29604.903540	-1222.903540
19621	23446.0	20769.153934	2676.846066
14965	46892.0	44276.790919	2615.209081
12321	40722.0	45462.870559	-4740.870559
6269	32084.0	35857.236859	-3773.236859

Figure 56: Predicted values

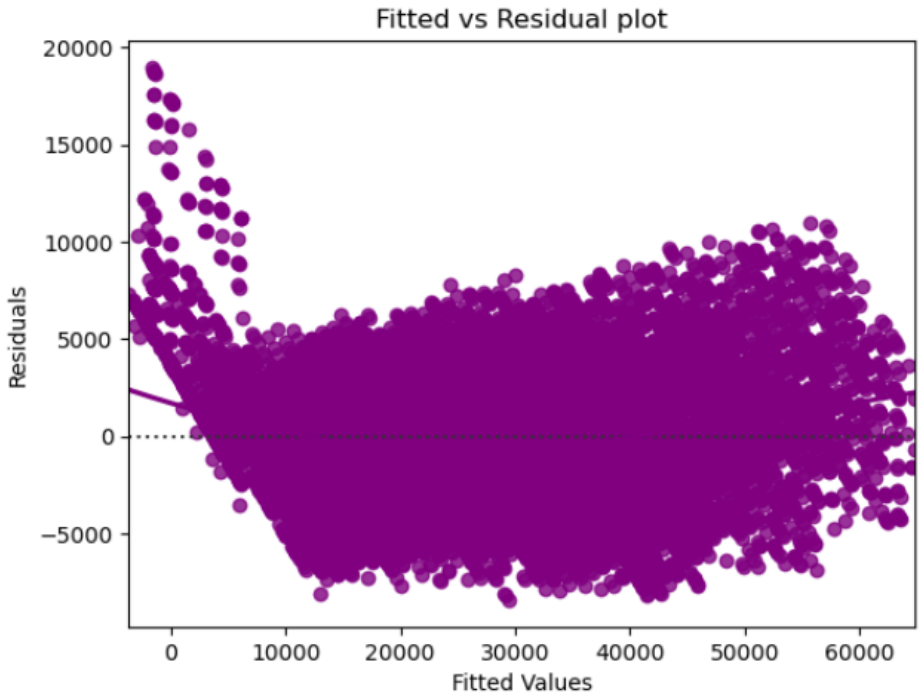


Figure 57: Residual Chart

- The scatter plot shows the distribution of residuals (errors) vs fitted values (predicted values).
- If there exist any pattern in this plot, we consider it as signs of non-linearity in the data and a pattern means that the model does not capture non-linear effects.
- **We see no pattern in the plot above. Hence, the assumptions of linearity and independence are satisfied.**

6. TEST FOR NORMALITY

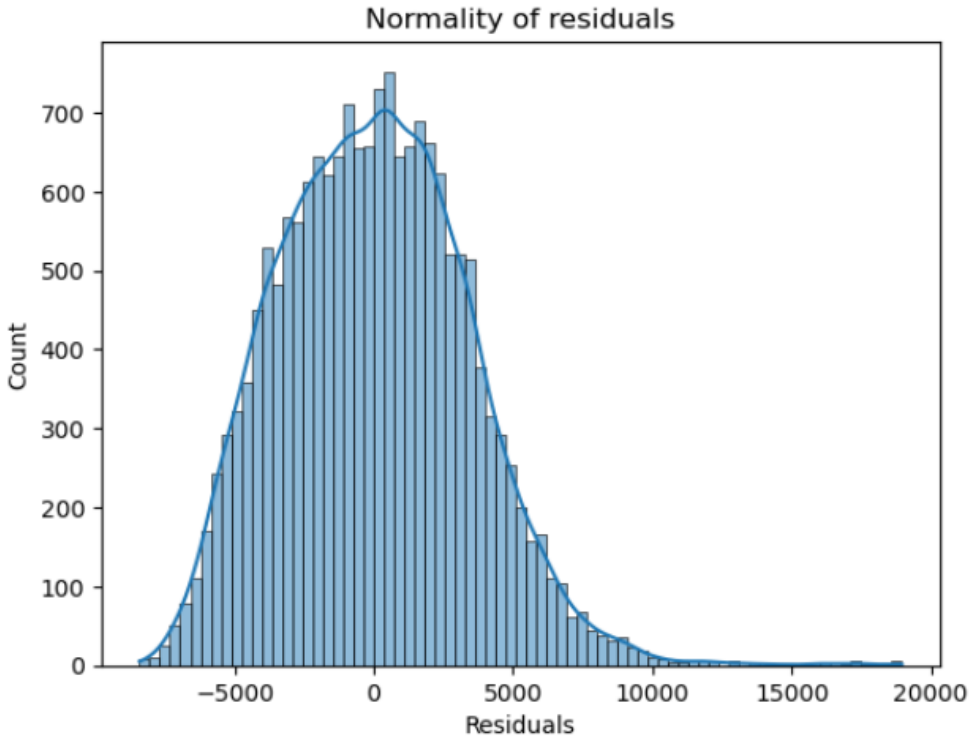


Figure 58: Normality of Residuals

- The histogram of residuals does have a bell shape.
- Let us check the Q-Q plot.

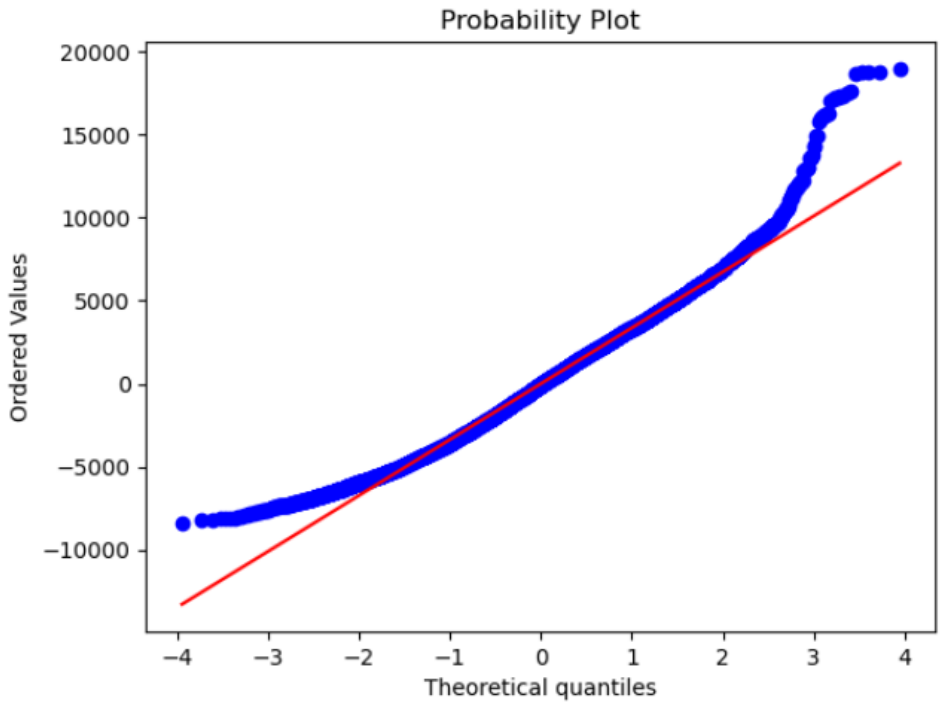


Figure 59: Q-Q Plot

7. **Shapiro Test**

```
ShapiroResult(statistic=0.987649019540111, pvalue=5.266053488552379e-36)
```

- Since p-value < 0.05, the residuals are not normal as per the Shapiro-Wilk test.
- Strictly speaking, the residuals are not normal.
- However, as an approximation, we can accept this distribution as close to being normal.
- **So, the assumption is satisfied.**

8. **F Statistic Test**

```
[('F statistic', 1.0007029442440367), ('p-value', 0.4869032216102235)]
```

Figure 60: F-Statistic

Since p-value > 0.05, we can say that the residuals are homoscedastic. So, this assumption is satisfied.

12.1.3 Model Performance on Test data

Test Performance

	RMSE	MAE	R-squared	Adj. R-squared	MAPE	Model
0	3340.182673	2711.95957	0.94479	0.944672	15.115261	OLS Model

Figure 61: Model Summary

- The model can explain ~94% of the variation in the data

12.1.4 LASSO Regression Model

LASSO Coefficients and Intercepts

```
Lasso Coefficients: [   0.          -12.26374286   23.49492635    2.81254998   53.72218469
   24.29975337 1490.45354718  174.38205263  -26.46086157 -450.99606236
  -32.33464418   -6.57755496   15.78393108   -6.23876337   23.62149231
 1198.54543901]
Intercept: -80115.19393698919
```

Figure 62: Model Coefficients

12.1.5 Model Performance on Training data

```
Training Performance
```

	RMSE	MAE	R-squared	Adj. R-squared	MAPE	Model
0	3383.989207	2726.856992	0.944539	0.944488	15.362929	LASSO Regression Model

Figure 63: Model Summary

12.1.6 Model Performance on Test data

```
Test Performance
```

	RMSE	MAE	R-squared	Adj. R-squared	MAPE	Model
0	3339.870964	2711.725453	0.9448	0.944682	15.11247	LASSO Regression Model

Figure 64: Model Summary

12.1.7 Ridge Regression Model

Ridge Coefficients and Intercepts

```
Ridge Coefficients: [   0.          -13.00049281   25.00009673    2.82287846   56.84577733
   26.80010695 1490.47528828  175.27379223  -29.4757048  -452.38385525
  -33.99088784   -6.76972836   20.9725242    -9.35072696   31.14397109
 1209.09990176]
Intercept: -80125.79032924659
```

Figure 65: Model Coefficients

12.1.8 Model Performance on Training data

```
Training Performance
```

	RMSE	MAE	R-squared	Adj. R-squared	MAPE	Model
0	3383.976847	2726.865727	0.94454	0.944489	15.364054	Ridge Regression Model

Figure 66: Model Summary

12.1.9 Model Performance on Test data

Test Performance

	RMSE	MAE	R-squared	Adj. R-squared	MAPE	Model
0	3340.181495	2711.959904	0.94479	0.944672	15.115224	Ridge Regression Model

Figure 67: Model Summary

12.1.10 Polynomial Regression Model

Polynomial Coefficients and Intercepts with Degree 2

```
Polynomial Coefficients: [-2.32620418e-12  3.05801649e+13 -1.49093017e+13  2.68806813e+13
  2.79347620e+12 -8.04821461e+11 -7.23866105e+10 -5.50793322e+11
 -9.51081988e+11  1.09306005e+11  1.63064589e+12 -5.15544525e+11
 -9.11311039e+11  4.10417096e+11 -3.43810583e+10 -1.55761462e+11
 -1.15743111e+11  4.03096959e+11  1.49093017e+13 -2.68806813e+13
 -2.79347620e+12  8.04821461e+11  7.23866105e+10  5.50793323e+11
  9.51081988e+11 -1.09306005e+11 -1.63064589e+12  5.15544525e+11
  9.11311038e+11 -4.10417096e+11  3.43810588e+10 -5.73185327e+09
  8.15123762e+10 -3.28271484e+00  7.18719482e+00 -1.38476562e+00
 -1.21018066e+01  3.53593750e+01 -6.12890625e+00  2.00186768e+01
 -9.99511719e-01 -8.35086060e+00  1.12084961e+00  1.85058594e+00
  2.94323120e+01  7.28900146e+00 -2.24451752e+01  2.42598114e+01
  3.11630707e+01 -1.45770264e+00  9.47769928e+00  1.33621063e+01
  6.02783203e-01  1.06763611e+01  2.66733246e+01  9.40368652e+00
 -3.17717056e+01  4.15069313e+01 -2.93419552e+01 -3.08427734e+01
  8.73083115e+00 -1.93857994e+01  1.10351562e-01  7.19604492e-01
 -3.16162109e+00  2.53906250e-01  2.92968750e-03  3.72070312e-01
  1.22070312e-02  1.11816406e+00 -1.55419922e+00  3.18237305e-01
  8.93554688e-01 -3.25756836e+00  1.22314453e-01 -2.58117065e+01
  6.38427734e-01 -2.49218750e+00 -1.55403137e+01 -4.87280884e+01
  1.19592590e+01 -9.64092064e+00  2.09856300e+01 -1.07066861e+02
  8.18051910e+01  3.93762474e+01  1.01526005e+02  5.08978882e+01
  2.55078125e+00  7.19604492e+00 -9.19572449e+00 -1.82461548e+00
 -3.76477661e+01 -9.71563721e+00 -6.86494751e+01 -2.24437714e+01
 -2.31682892e+01 -1.23584734e+02  7.03515625e+00 -5.39062500e+00
  2.27148438e+00 -1.48691406e+01  1.50683594e+00  2.69628906e+00
 -5.94482422e-01 -6.99267578e+00 -3.97753906e+00  8.74555664e+01
  9.59450073e+01 -6.37927246e+01 -5.21328430e+01 -2.62469482e+00
 -2.46665955e-01 -4.52156830e+00 -2.71109009e+00 -3.10994453e+01
 -1.57767212e+02 -1.53644409e+01 -4.95735168e+00  2.09503174e-01
  6.29699707e-01 -7.11988525e+01  1.28529510e+01 -4.47697067e+01
 -5.86489677e+01  9.05541382e+01 -3.68533936e+01  3.62471619e+01
 -3.15697327e+01  1.24545135e+01  1.76093903e+01  2.21622314e+01
  5.27673340e+00  1.81825867e+01 -7.32446289e+00 -5.81988525e+01
  2.96691895e-01  1.28589935e+01 -4.63595581e+00  1.79979248e+01
  4.13490982e+01  1.64613190e+01 -2.71899223e+01 -3.36017838e+01
  5.06747246e+01  1.24331081e+02 -2.99902935e+01 -5.62976227e+01
  2.47227287e+01  1.14948166e+02  1.61493316e+11  1.54255295e+01
  3.42307299e+10]
Intercept: -30983261909551.254
```

Figure 68: Model Coefficients

12.1.11 Model Performance on Training data

```
Training Performance
```

	RMSE	MAE	R-squared	Adj. R-squared	MAPE	Model
0	3211.688318	2608.852823	0.950043	0.949997	13.984243	Polynomial Regression Model

Figure 69: Model Summary

12.1.12 Model Performance on Test data

```
Test Performance
```

	RMSE	MAE	R-squared	Adj. R-squared	MAPE	Model
0	3200.639262	2612.016973	0.949306	0.949198	13.906082	Polynomial Regression Model

Figure 70: Model Summary

12.1.13 Model Performance Comparison on Training data between Parametric Models

```
Training performance comparison:
```

	RMSE	MAE	R-squared	Adj. R-squared	MAPE	Model
0	3383.976844	2726.866303	0.944540	0.944489	15.364108	OLS Model
1	3383.989207	2726.856992	0.944539	0.944488	15.362929	LASSO Regression Model
2	3340.181495	2711.959904	0.944790	0.944672	15.115224	Ridge Regression Model
3	3211.688318	2608.852823	0.950043	0.949997	13.984243	Polynomial Regression Model

Figure 71: Model Summary

12.1.14 Model Performance Comparison on Test data between Parametric Models

Testing performance comparison:

	RMSE	MAE	R-squared	Adj. R-squared	MAPE	Model
0	3340.182673	2711.959570	0.944790	0.944672	15.115261	OLS Model
1	3339.870964	2711.725453	0.944800	0.944682	15.112470	LASSO Regression Model
2	3383.976847	2726.865727	0.944540	0.944489	15.364054	Ridge Regression Model
3	3200.639262	2612.016973	0.949306	0.949198	13.906082	Polynomial Regression Model

Figure 72: Model Summary

Polynomial Regression Model with Degree 2 has Lesser MAPE and Higher Adjusted R Squared compared to other Parametric Models.

12.2 Non-Parametric Models

Data is split into 3 sets, Train, Validation, Test.

```
Number of rows in train data = 15000
Number of rows in validation data = 5000
Number of rows in test data = 5000
```

Figure 73: Data Split

12.2.1 Model Performance Comparison on Training and Validation data

```
Training Performance:

Random forest:
          RMSE         MAE  R-squared  Adj. R-squared      MAPE
0  1168.241102  926.231351   0.993361        0.993354  4.543973

XGB:
          RMSE          MAE  R-squared  Adj. R-squared      MAPE
0  2307.866801  1821.500487   0.974091        0.974063  9.071529

Adaboost:
          RMSE          MAE  R-squared  Adj. R-squared       MAPE
0  3250.350122  2690.963374   0.948608        0.948553  15.848071

SVR:
           RMSE          MAE  R-squared  Adj. R-squared      MAPE
0  14058.819429  11595.92706    0.03854        0.037514  78.59003


Validation Performance:

Random forest:
          RMSE          MAE  R-squared  Adj. R-squared       MAPE
0  3124.925289  2502.941944   0.952283        0.952129  12.339513

XGB:
          RMSE          MAE  R-squared  Adj. R-squared       MAPE
0  3146.491594  2531.833429   0.951622        0.951466  12.594991

Adaboost:
          RMSE          MAE  R-squared  Adj. R-squared       MAPE
0  3280.273472  2705.700624    0.94742        0.947251  15.927634

SVR:
          RMSE           MAE  R-squared  Adj. R-squared       MAPE
0  14025.58842  11577.663506   0.038741        0.035655  78.701528
```

Figure 74: Model Summary

12.2.2 Model Performance Differences on Training and Validation data

```
Training and Validation Performance Difference:

Random forest:
  RMSE  - Train: 1168.24, Validation: 3124.93, Diff: -1956.68
  MAE   - Train: 926.23, Validation: 2502.94, Diff: -1576.71
  R²    - Train: 0.99, Validation: 0.95, Diff: 0.04
  Adj R²- Train: 0.99, Validation: 0.95, Diff: 0.04
  MAPE  - Train: 4.54%, Validation: 12.34%, Diff: -7.80%

XGB:
  RMSE  - Train: 2307.87, Validation: 3146.49, Diff: -838.62
  MAE   - Train: 1821.50, Validation: 2531.83, Diff: -710.33
  R²    - Train: 0.97, Validation: 0.95, Diff: 0.02
  Adj R²- Train: 0.97, Validation: 0.95, Diff: 0.02
  MAPE  - Train: 9.07%, Validation: 12.59%, Diff: -3.52%

Adaboost:
  RMSE  - Train: 3250.35, Validation: 3280.27, Diff: -29.92
  MAE   - Train: 2690.96, Validation: 2705.70, Diff: -14.74
  R²    - Train: 0.95, Validation: 0.95, Diff: 0.00
  Adj R²- Train: 0.95, Validation: 0.95, Diff: 0.00
  MAPE  - Train: 15.85%, Validation: 15.93%, Diff: -0.08%

SVR:
  RMSE  - Train: 14058.82, Validation: 14025.59, Diff: 33.23
  MAE   - Train: 11595.93, Validation: 11577.66, Diff: 18.26
  R²    - Train: 0.04, Validation: 0.04, Diff: -0.00
  Adj R²- Train: 0.04, Validation: 0.04, Diff: 0.00
  MAPE  - Train: 78.59%, Validation: 78.70%, Diff: -0.11%
```

Figure 75:Model Summary

12.2.3 All Model Performance Differences on Training data

Model	RMSE	MAE	R-squared	Adj. R-squared	MAPE
Random forest	1168.241102	926.231351	0.993361	0.993354	4.543973
XGB	2307.866801	1821.500487	0.974091	0.974063	9.071529
Adaboost	3250.350122	2690.963374	0.948608	0.948553	15.848071
SVR	14058.819429	11595.927060	0.038540	0.037514	78.590030

Figure 76: Model Summary

Models like Random Forest, XGB, AdaBoost are good with MAPE but the models might be overfitted. This raises the need for Hyperparameter Tuning all the models.

After initial model runs, we undertook systematic tuning efforts for non-parametric models to enhance accuracy and prevent overfitting:

12.2.4 Tuned Random Forest Regression Model Performance on Train and Validation data

```
 Final Scores for Best Estimator:
Training Set:
          RMSE         MAE  R-squared  Adj. R-squared      MAPE  \
0  1227.742216  983.113595   0.992668         0.99266  5.137801

                Model
0  Tuned RandomForest

Validation Set:
          RMSE          MAE  R-squared  Adj. R-squared       MAPE
0  3247.031591  2608.949572   0.948481        0.948315  13.687654
CPU times: total: 10.6 s
Wall time: 1min 32s
```

Figure 77: Model Summary

12.2.5 Tuned XG Boost Regression Model Performance on Train and Validation data

```
Best Parameters Found:
{'subsample': 0.8, 'n_estimators': 175, 'max_depth': 3, 'learning_rate': 0.05}

 Performance Metrics – Best XGBoost Estimator
Training:
          RMSE          MAE  R-squared  Adj. R-squared       MAPE  \
0  2974.037953  2399.201729   0.956974        0.956929  11.939725

           Model
0  Tuned XGBoost

Validation:
          RMSE          MAE  R-squared  Adj. R-squared      MAPE
0  3004.052406  2418.236994   0.955903        0.955761  11.90582
```

Figure 78: Model Summary

12.2.6 Tuned Adaboost Regressor Model Performance on Train and Validation data

```
Best Parameters:
{'n_estimators': 175, 'loss': 'exponential', 'learning_rate': 0.05}

 Performance Metrics – Best AdaBoost Model
Training:
          RMSE          MAE  R-squared  Adj. R-squared       MAPE  \
0  3194.328316  2594.425703   0.950364        0.950311  13.977834

            Model
0  Tuned Adaboost

Validation:
          RMSE          MAE  R-squared  Adj. R-squared       MAPE
0  3228.724341  2611.744489    0.94906        0.948896  14.041136
```

Figure 79: Model Summary

12.2.7 Tuned SVR Regressor Model Performance on Train and Validation data

```
Best Parameters:
{'svr__kernel': 'linear', 'svr__gamma': 'scale', 'svr__epsilon': 0.1, 'svr__C': 10}

 Performance Metrics – Best SVR Model
Training:
          RMSE          MAE  R-squared  Adj. R-squared       MAPE      Model
0  3399.810658  2723.705345   0.943773        0.943713  15.113603  Tuned SVR

Validation:
          RMSE          MAE  R-squared  Adj. R-squared       MAPE
0  3414.964466  2723.795565   0.943014        0.942831  15.062678
```

Figure 80: Model Summary

12.2.8 Model Performance Comparison on Training data between Non-Parametric Models

Model	RMSE	MAE	R-squared	Adj. R-squared	MAPE
Tuned RandomForest	1227.742216	983.113595	0.992668	0.992660	5.137801
Tuned XGBoost	2974.037953	2399.201729	0.956974	0.956929	11.939725
Tuned Adaboost	3194.328316	2594.425703	0.950364	0.950311	13.977834
Tuned SVR	3399.810658	2723.705345	0.943773	0.943713	15.113603

Figure 81: Model Summary

12.2.9 All Model Performance Comparison on Training data between Non-Parametric Models

Model	RMSE	MAE	R-squared	Adj. R-squared	MAPE
Random forest	1168.241102	926.231351	0.993361	0.993354	4.543973
XGB	2307.866801	1821.500487	0.974091	0.974063	9.071529
Adaboost	3250.350122	2690.963374	0.948608	0.948553	15.848071
SVR	14058.819429	11595.927060	0.038540	0.037514	78.590030
Tuned RandomForest	1227.742216	983.113595	0.992668	0.992660	5.137801
Tuned XGBoost	2974.037953	2399.201729	0.956974	0.956929	11.939725
Tuned Adaboost	3194.328316	2594.425703	0.950364	0.950311	13.977834
Tuned SVR	3399.810658	2723.705345	0.943773	0.943713	15.113603

Figure 82: Model Summary

12.2.10 Model Performance Comparison on Test data between Non-Parametric Models

Model	RMSE	MAE	R-squared	Adj. R-squared	MAPE
Tuned RandomForest	1227.742216	983.113595	0.992668	0.992660	5.137801
Tuned XGBoost	2974.037953	2399.201729	0.956974	0.956929	11.939725
Tuned Adaboost	3194.328316	2594.425703	0.950364	0.950311	13.977834
Tuned SVR	3399.810658	2723.705345	0.943773	0.943713	15.113603

Figure 83: Model Summary

12.2.11 All Model Performance Comparison on Test data between All Models

Model	RMSE	MAE	R-squared	Adj. R-squared	MAPE
OLS Model	3383.976844	2726.866303	0.944540	0.944489	15.364108
LASSO Regression Model	3383.989207	2726.856992	0.944539	0.944488	15.362929
Ridge Regression Model	3340.181495	2711.959904	0.944790	0.944672	15.115224
Polynomial Regression Model	3211.688318	2608.852823	0.950043	0.949997	13.984243
Tuned RandomForest	1227.742216	983.113595	0.992668	0.992660	5.137801
Tuned XGBoost	2974.037953	2399.201729	0.956974	0.956929	11.939725
Tuned Adaboost	3194.328316	2594.425703	0.950364	0.950311	13.977834
Tuned SVR	3399.810658	2723.705345	0.943773	0.943713	15.113603

Figure 84: Model Summary

12.2.12 All Model Performance Comparison on Train data between All Models Sorted

Model	RMSE	MAE	R-squared	Adj. R-squared	MAPE
Tuned RandomForest	1227.742216	983.113595	0.992668	0.992660	5.137801
Tuned XGBoost	2974.037953	2399.201729	0.956974	0.956929	11.939725
Tuned Adaboost	3194.328316	2594.425703	0.950364	0.950311	13.977834
Polynomial Regression Model	3211.688318	2608.852823	0.950043	0.949997	13.984243
Tuned SVR	3399.810658	2723.705345	0.943773	0.943713	15.113603
Ridge Regression Model	3340.181495	2711.959904	0.944790	0.944672	15.115224
LASSO Regression Model	3383.989207	2726.856992	0.944539	0.944488	15.362929
OLS Model	3383.976844	2726.866303	0.944540	0.944489	15.364108

Figure 85: Model Summary

12.2.13 All Model Performance Comparison on Test data between All Models Sorted

Model	RMSE	MAE	R-squared	Adj. R-squared	MAPE
Tuned XGBoost	3108.266238	2510.930901	0.952741	0.952589	12.257640
Tuned Random Forest	3333.187473	2694.652893	0.945654	0.945480	13.758945
Polynomial Regression Model	3200.639262	2612.016973	0.949306	0.949198	13.906082
Tuned Adaboost	3302.679856	2678.215204	0.946644	0.946473	14.240940
LASSO Regression Model	3339.870964	2711.725453	0.944800	0.944682	15.112470
OLS Model	3340.182673	2711.959570	0.944790	0.944672	15.115261
Ridge Regression Model	3383.976847	2726.865727	0.944540	0.944489	15.364054
Tuned SVR	3486.833863	2800.861362	0.940528	0.940337	15.443473

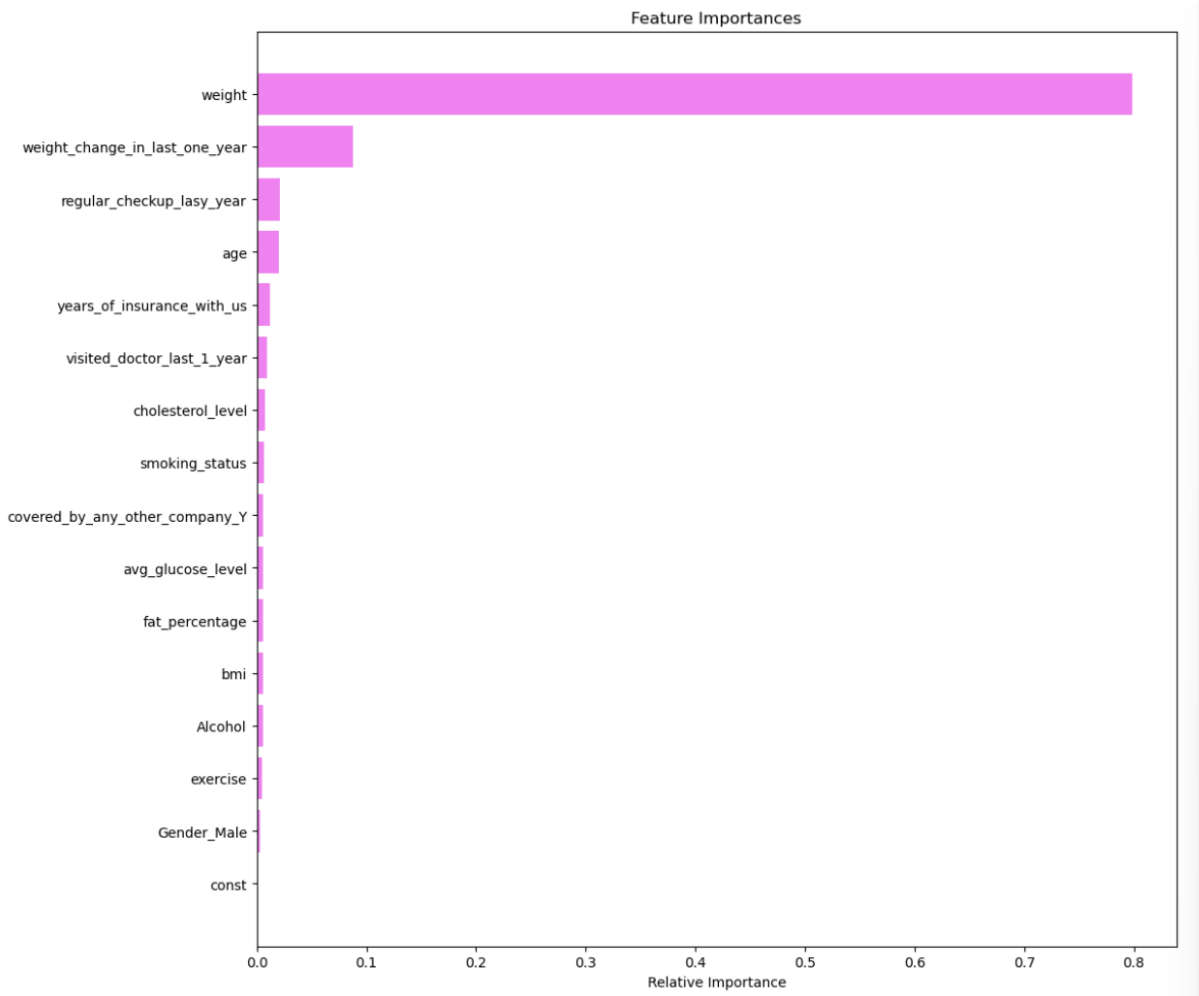
Figure 86: Model Summary

All models have given MAPE less than 15% on test data compared with Train data

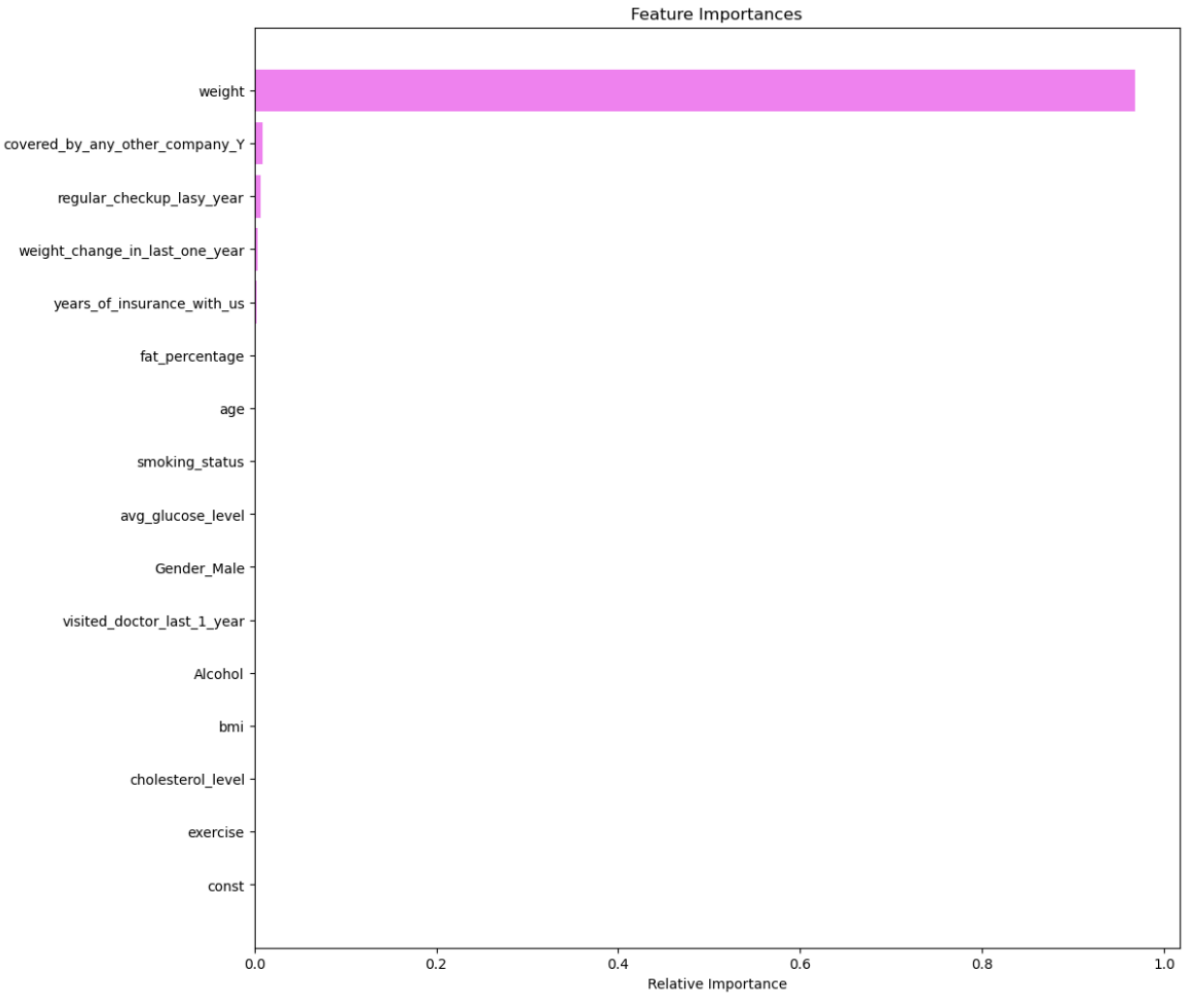
This performance is in line with what we achieved with this model on the train and validation sets

So, picking top 2 models to check feature importances

12.4 Feature Importance of Tuned Random Forest Model



12.5 Feature Importance of Tuned XG Boost Model



13.0 Interpretation of the Most Optimum model

Tuned XG Boost:

- The Tuned XG Boost model stands out as the most accurate and business-aligned choice in lineup. With a MAPE of just 12.26%, it offers exceptionally tight forecasting performance.
- Its R-squared value of 0.9527 confirms that it captures over 95% of the variance in the target variable, signifying strong feature.
- The model also posted the lowest RMSE and MAE, smoother performance across diverse customer profiles.
- In terms of feature behaviour, XG Boost places overwhelming emphasis on “weight”. Other features like coverage by another insurance company and regular health checkups register minor but intuitive signals, highlighting behavioural and contextual flags.

Tuned Random Forest:

- The Tuned XG Boost model stands out as the slightly lesser accurate than Tuned XG Boost model. With a MAPE of just 13.76%, it offers exceptionally tight forecasting performance.
- Its R-squared value of 0.9457 confirms that it captures over 94% of the variance in the target variable, this model still delivers highly dependable performance — just a notch below XG Boost.
- The slightly higher RMSE and MAE suggest broader prediction intervals but remain well within acceptable thresholds.
- In terms of feature behaviour, Random Forest truly shines in its clarity of feature importances. Not only does it reaffirm **weight** as the primary driver ~80% importance, but it also surfaces **weight change in the last year** as a meaningful secondary signal ~10%. Other features like **checkups**, **age**, **tenure**, and **lifestyle indicators** contribute minimally

14.0 Validation Approach

The model validation process went beyond just accuracy. We used a **multi-step evaluation strategy**:

1. Train-Test Split & Cross-Validation

- Data split into **Train (70%)** and **Test (30%)** for initial validation.
- For non-parametric models (Random Forest, XGBoost, etc.), **Train-Validation-Test split** was used to tune hyperparameters and avoid overfitting.

2. Performance Metrics

- Instead of only accuracy, we focused on metrics that reflect **business needs** for cost prediction:
    - **R^2 (Coefficient of Determination):** To measure variance explained by the model.
        - Tuned XGBoost achieved **R^2 = 0.9527** (very high explanatory power).
    - **MAPE (Mean Absolute Percentage Error):** Business-friendly metric for forecasting errors in percentage terms.
        - Tuned XGBoost achieved **MAPE = 12.26%**.
    - **RMSE & MAE:** To assess absolute error magnitude and model stability.

3. Model Assumptions & Diagnostics (for Parametric Models)

- **Multicollinearity Check:** Variance Inflation Factor (VIF) calculated; variables with VIF > 5 were reviewed.
- **Residual Analysis:** Checked for normality (Q-Q plots), independence (residual plots), and homoscedasticity.
- **Shapiro-Wilk Test:** Verified normality of residuals (with approximation accepted).

4. Overfitting Control

- Hyperparameter tuning performed for non-parametric models using **Grid Search** and **Cross-Validation**.
- Compared **Train vs Test performance** to ensure generalization (difference within acceptable range).

5. Feature Importance Analysis

- Post-validation, analysed feature contributions using **XGBoost's feature importance scores** to ensure the model's logic aligns with domain knowledge (weight being top predictor makes sense).

Impact:
Hyperparameter tuning significantly enhanced prediction accuracy, reduced error margins, and stabilized model performance for real-world deployment.

15.0 Insights and Recommendations

15.1 Insights from Analysis

1. Weight (~95% importance)

Direction of Impact: Positive.

Interpretation: Higher body weight generally reflects higher BMI and fat %, both linked to increased cardio-metabolic risk (e.g., diabetes, heart disease).

Business Impact: Individuals with higher weight are more likely to incur higher medical expenses, leading to higher insurance costs.

2. Coverage by Another Insurance Company (~1.5% importance)

Direction of Impact: Mixed.

Interpretation: Having additional coverage could mean:

Lower actual utilization of this policy (cost dilution), OR

Higher baseline risk prompting individuals to seek multiple coverages.

Business Impact: While the direct correlation with cost is minor, these variable signals customer risk perception and can influence premium structuring or cross-sell opportunities.

3. Regular Health Checkups (~1.5% importance)

Direction of Impact: Positive.

Interpretation:

Higher checkup frequency often means active health monitoring — but in insurance pricing, it may also flag existing chronic conditions requiring continuous management.

Preventive care can lower long-term risks, but short-term insurance cost may be higher due to increased utilization.

Business Impact:

Can serve as a leading indicator for proactive care segments.

May justify wellness incentives or discounts to encourage preventive habits while managing claims.

4. Other Features (Combined ~2%)

BMI, Avg Glucose Level, Fat %, Age, Smoking, Cholesterol, Exercise, Alcohol, Weight Change, Gender, Tenure

General Impact:

These contribute marginally in isolation but can have stronger predictive power when interacting with top features (e.g., BMI + Age, Glucose + Weight).

Provide context to refine customer health profiles, underwriting, and segmentation.

15.2 Recommendations

Weight Management Programs (Top Predictor – 95% importance)

1. Launch targeted wellness programs focusing on **weight reduction** and **healthy BMI maintenance**.
2. Offer **premium discounts** or **cashback incentives** for customers achieving set weight or fitness targets.
3. Partner with gyms, nutritionists, and wellness apps for **corporate health packages**.

Engagement for Low-Touch Segments (Cluster 1 – Silent Risk)

1. Create **outreach campaigns** for customers with **few/no checkups** and low activity levels.
2. Offer free annual screenings or preventive care vouchers to encourage early diagnosis.
3. Use **mobile nudges** and gamified health challenges to boost participation.

Customized Plans for Dual-Coverage Holders

1. For customers with **coverage from another insurance company**, design **add-on or supplementary plans** to capture additional engagement.
2. Position these as "gap coverage" for risks not fully covered by their other policies.

Incentivizing Preventive Healthcare

1. While frequent checkups indicate higher short-term costs, they can lower **long-term claim severity**.
2. Introduce **wellness rewards** for preventive screenings, chronic condition management programs, and health coaching.

Address Data Ambiguity (Cluster 3)

1. High rates of **unknown smoking status** and other missing lifestyle details can hide risk.
2. Improve **data capture** at onboarding and through periodic surveys.
3. Consider **underwriting review** or predictive risk scoring for incomplete profiles.

Promote Active Lifestyles

1. Leverage step-count and exercise-level data to reward customers achieving **activity milestones**.
2. Partner with wearable tech companies to link **activity tracking with premium discounts**.